

Chapter 44. Barotrauma and Bronchopleural Fistula

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Barotrauma and Bronchopleural Fistula: Introduction

This chapter considers the possible origins of extraalveolar air in ventilated patients and then reviews the manifestations and most frequent clinical settings of its various forms. After a discussion of general principles of management and a review of reported therapeutic approaches, it presents logical steps for prevention in susceptible patients and points out important gaps in the existing evidence base pertaining to this important topic. Although all the clinical forms of barotrauma are touched upon, most attention is devoted to those that pose a threat to life.

Only overt extraalveolar air is covered in this chapter. The reader is referred to [Chapter 42](#) for a discussion of lung damage at the tissue or subcellular level related to mechanical lung distension and the application of positive pressure to the airways. Because data from laboratory studies are covered extensively in that discussion, this chapter deals primarily with barotrauma as a complication in patients, referring mainly to the adult clinical literature. As will be apparent, although the latter is replete with anecdotes and observational reports, this focus primarily on human data means that the evidence base in terms of prospective studies and “hard data” available to the clinician is remarkably limited. Spontaneous pneumothorax and other forms of extraalveolar air encountered in patients who are not intubated or receiving mechanical ventilation are not dealt with extensively here, nor is decompression-related barotrauma or bronchopleural fistula complicating lung resection.

Definitions

Pneumothorax, subcutaneous emphysema, and other clinical forms of extraalveolar air occurring in association with mechanical ventilation are commonly referred to as *barotrauma*. This term is doubly unfortunate, *trauma* connoting iatrogenic injury and *baro* implying that it is pressure, rather than volume, shear force, or some other factor that produces it. In fact, these implications of both roots of the word are probably incorrect, and the expression *ventilator-associated extraalveolar air* would be technically more appropriate. Similarly, the term *bronchopleural air leak* would be more accurate than *bronchopleural fistula*, because of the implications of inflammation and suppuration associated with the word *fistula* in surgical and other settings. Like *barotrauma*, however, the latter is so ingrained in clinical usage that change is unlikely, and the more familiar terms are used in this chapter, as elsewhere in this book.

Although extraalveolar air appearing during ventilator support may not be caused by the ventilator itself, the expression *ventilator-induced lung injury* would seem as applicable to clinical barotrauma as to parenchymal damage associated with mechanical stretch and overdistension. Again, however, owing more to convention than to logical etymology, the term *ventilator-induced lung injury* is generally reserved for the latter, in this book and in the broader literature.

Usually, barotrauma in mechanically ventilated patients is automatically assumed to be a complication of mechanical ventilation. As with nosocomial pneumonia, however, ventilatory muscle dysfunction, and most of the other adverse developments to which ventilated patients are prone, whether the ventilator per se is responsible is usually unclear,

because patients ill enough to require endotracheal intubation, supplemental [oxygen](#), positive-pressure ventilation, and management in an intensive care unit (ICU) tend to have numerous other predispositions to such complications. Each of these things would be better thought of as complications *associated with*, rather than *of*, mechanical ventilation.^{1,2}

Pathophysiology

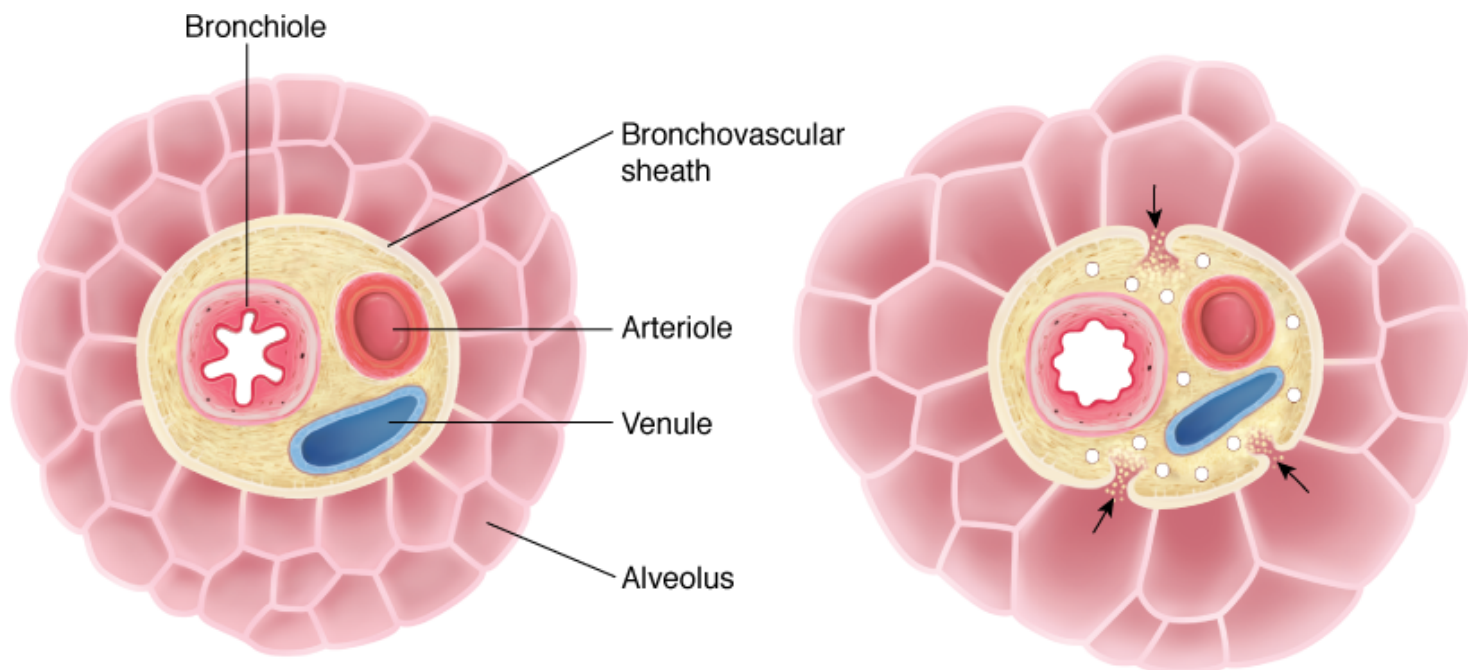
Mechanism of Alveolar Disruption

Barotrauma in a patient receiving positive-pressure ventilation has a number of potential causes ([Table 44-1](#)).³ Most often extraalveolar air during mechanical ventilation results from overdistension of alveoli and rupture of their walls down a pressure gradient from airspace into bronchovascular sheath, as illustrated in [Figure 44-1](#).⁴ This mechanism was worked out many years ago in elegant animal experiments by C.C. and M.T. Macklin:^{5,6} “... pulmonic interstitial emphysema and its sequelae—air in the mediastinum, peritoneal cavity, subcutaneous tissues, and pleural cavity—are present in many conditions; differing widely in their cause, their clinical manifestations and their seriousness. All of these conditions, however, have a single common factor, a gradient of pressure between the alveoli and vessel sheath, and hence an opportunity for air to gain access to the interstitial tissues of the lung.”⁶

Table 44-1: Possible Origins of Extraalveolar Air

Upper respiratory tract
Fractures of facial bones, mandible, etc.
Other traumatic mucosal disruption
Dental extractions and other oral surgical procedures
Intrathoracic airways
Airway rupture or laceration in blunt or penetrating chest trauma
Complications of intubation or airway instrumentation
Foreign body in upper or lower airways
Bronchoscopy-related procedures (e.g., transbronchial biopsy, bronchial brushing, or transbronchial needle aspiration)
Lung parenchyma
Penetrating trauma
Surgical procedures
Diagnostic procedures as above
Thoracentesis
Percutaneous needle aspiration or biopsy
Alveolar rupture
Gastrointestinal tract
Perforation of esophagus or abdominal viscus
Infection with gas-producing organisms
Pleural empyema
Acute mediastinitis
Necrotizing fasciitis or other soft-tissue infection
Exogenous source (air from outside the body)
Penetrating trauma
Thoracentesis or closed pleural biopsy
Surgical procedures (e.g., chest tube insertion, tracheotomy, or mediastinoscopy)

Figure 44-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Mechanism of alveolar rupture during mechanical ventilation. Pressures between adjacent alveoli equalize rapidly, but, especially in the presence of high alveolar volume, increased alveolar pressure, in comparison with that in the adjacent bronchovascular sheath, establishes a pressure gradient that may result in rupture of the alveolar wall, allowing passage of air into the interstitial tissue of the bronchovascular sheath. (Used, with permission, from Maunder et al.⁴)

According to the work of the Macklins,^{5,6} the basic requirement for alveolar rupture is the presence of a pressure gradient between the alveoli and their surrounding structures. Intervalveolar walls are probably not susceptible to such pressure gradients, as they are very thin and the pressures between adjacent alveoli are probably equal. Any sudden increase in alveolar pressure (or presumably a fall in perivascular interstitial pressure), however, may establish a gradient sufficiently large to disrupt alveolar walls at their bases (see Fig. 44-1), introducing air into the pulmonary interstitium. Alveolar overdistension would tend to thin and stretch the alveolar membranes, facilitating rupture once the pressure gradient was established. This mechanism can readily be understood in the case of extraalveolar air following sudden deceleration injury, such as falling into water from a height,⁷ but it likely is the main means of alveolar rupture in the majority of other clinical settings as well.^{3,4,8}

Spread of extraalveolar air via bronchovascular sheaths after alveolar rupture was dramatically demonstrated by Jamadar et al when pneumomediastinum developed in a patient undergoing liquid ventilation.⁹ After the liquid medium was removed from the bronchial tree and alveoli, computed tomography of the chest demonstrated persistence of the radiopaque perfluorocarbon fluid in the bronchovascular sheaths, confirming the mechanism elucidated a half-century earlier by the Macklins.

Other Possible Sources of Extraalveolar Air

Beyond clinical barotrauma that occurs as a direct complication of mechanical ventilation, it is important to be cognizant of other possible sources not related to the ventilator itself (Table 44-2).^{3,8} Distinguishing these causes from barotrauma secondary to mechanical ventilation itself may be challenging in some circumstances, particularly when adequate clinical history is lacking. Pneumothorax appearing within the first few hours following initiation of mechanical ventilation may

be the result of previous trauma, overinflation during manual ventilation, or procedures such as central line attempts before intubation. Even in patients already receiving ventilator support, the appearance of extraalveolar air may represent “pseudobarotrauma”—external trauma to the lung from attempts at placing central venous access or various endoscopic procedures—rather than “spontaneous” alveolar rupture.

Table 44-2: Possible Mechanisms for “Barotrauma” in Mechanically Ventilated Patients

Airway disruption or alveolar rupture before initiation of ventilatory support
Trauma (penetrating or blunt)
Resuscitation (mouth-to-mouth or manual ventilation)
Airway laceration or perforation during intubation attempts
Attempted central line placement (e.g., via the internal jugular or subclavian route)
Biopsy or surgical procedure
Direct laceration of visceral pleura or airway during mechanical ventilation (“pseudobarotrauma”)
Central line placement
Thoracentesis or chest tube placement
Transbronchial biopsy or bronchial brushing
“Spontaneous” alveolar rupture
Manifestation of a primary disease process
Complication of ventilator-associated pneumonia or sepsis
Inadvertent alveolar overdistension (e.g., right main bronchus intubation or manual ventilation)
Related to ventilator management per se (e.g., tidal volume, positive end-expiratory pressure, recruitment maneuvers, or breath-stacking)

Air originating in the upper respiratory tract can dissect downward from the head and neck and produce subcutaneous emphysema, pneumomediastinum, and conceivably even pneumothorax. During mechanical ventilation, this could occur in the presence of negative intrathoracic pressure, as with vigorous efforts against a partially occluded airway or in severe patient-ventilator dyssynchrony. Reported causes of pneumomediastinum from air dissection from above include facial or mandibular fractures,¹⁰ retropharyngeal abscess,¹¹ and dental extractions, particularly if these involve the lower molar teeth and air-turbine drilling.¹²

Mediastinal air and other barotrauma can also originate in the intrathoracic airways, as after blunt or penetrating chest trauma,^{13,14} as a complication of bronchoscopic procedures, during percutaneous dilatational tracheotomy,¹⁵ or from perforation by a foreign body. Injury to the upper airway or laceration of the posterior membranous trachea may occur during attempted endotracheal intubation, particularly when a stylet is used, and this has been associated with clinical barotrauma in numerous reports.^{16–20}

Endoscopic procedures^{21–23} and other interventions may rupture or perforate the esophagus, providing access for air into the mediastinum. In many instances, but not always, signs of mediastinitis accompany the appearance of pneumomediastinum following esophageal procedures.⁸ Air in the mediastinum is also a characteristic feature of Boerhaave syndrome,²⁴ and the typical history of vigorous retching after a large meal may be absent.

Most often the origin of clinical barotrauma is the lung parenchyma, but even then the mechanism may be other than “spontaneous” alveolar rupture. Other possibilities include penetrating trauma that lacerates pulmonary tissue, surgical procedures involving the intrathoracic structures, bronchoscopic procedures, and transthoracic needle aspiration. When acute mediastinitis or pleural empyema involves gas-producing organisms, palpable or radiographically detectable extraalveolar air may result; gas arising from soft-tissue infections such as clostridial gangrene may also be confused with barotrauma.

Finally, exogenous air may find its way into the pleural space, subcutaneous tissues, or mediastinum. Most commonly this occurs during or following thoracentesis, chest tube insertion, tracheotomy,²⁵ or mediastinoscopy. Rarely, air may enter the soft tissues following cutaneous injury to an extremity²⁶ or elsewhere in the body, producing subcutaneous emphysema that could be confused with barotrauma.

Airway Pressure versus Alveolar Distension

For many years, a primary goal of mechanical ventilation was to limit peak airway pressures, based on the belief that high peak airway pressures were a primary mediator of barotrauma.^{27–31} The desire to avoid high peak airway pressures was a main driver in the introduction of pressure-targeted modes of ventilation and the different forms of high-frequency ventilation.³ As discussed in [Chapter 42](#), however, a large body of evidence now supports the conclusion that it is excessive volume, not airway pressure per se, and especially not peak airway pressure measured outside the patient, that primarily determines the occurrence and severity of ventilator-induced lung injury.

Peak airway pressure during mechanical ventilation can be influenced by a number of factors ([Table 44-3](#)), most of which do not affect (or reflect) alveolar volume. This can readily be understood by envisioning the various potential sources for high peak airway pressure as sensed at the ventilator’s manometer. Increased resistance to airflow in the ventilator circuit or endotracheal tube, or in the patient’s conducting airways, will be reflected in an increased peak inspiratory airway pressure. Likewise, an increase in chest wall pressure, as with coughing, chest strapping, or massive generalized edema, will increase peak airway pressure without increasing transalveolar pressure or alveolar volume. Even increased pressure at the alveolar level should not cause disruption of alveolar walls unless this pressure is unevenly distributed between adjacent alveoli or between alveoli and the adjoining bronchovascular sheath.

Table 44-3: Possible Causes of High Peak Airway Pressure during Mechanical Ventilation

High inspiratory flow (increases proximally measured peak airway pressure at any given total airway resistance)
High resistance in the ventilator circuit or endotracheal tube
Inspissated secretions
Smaller-than-usual endotracheal or tracheostomy tube
Kinking
Obstructed distal orifice
High resistance in the patient's airways
Bronchospasm
Secretions
Airway edema
Neoplasm, stenosis, or foreign body in trachea or main bronchus
Alveolar overdistension
Decreased compliance of the lung parenchyma
High pleural or transthoracic pressure
Coughing
Agitation, shivering, or seizures
Splinting as a consequence of chest wall pain
Pneumothorax or pleural effusion
Ascites or abdominal packs or binding
Head-down position
Bandages, casts, or restraints
Other causes of intraabdominal hypertension or reduced chest wall compliance

This conclusion becomes obvious when one considers the common activities that markedly raise peak airway pressure without causing alveolar rupture.³ Normal persons routinely generate peak airway pressures up to 200 cm H₂O or more during a cough or sneeze.³² In a study of fifty-six normal male subjects ranging in age from 6 to 64 years, Cook et al³³ found their mean individual maximum peak expiratory airway pressures to be 237 ± 45 cm H₂O. In that study, the subjects sustained these pressures for at least 1 to 2 seconds, and even longer periods of sustained high peak airway pressures can be maintained during the Valsalva maneuver, during which pressures as high as 200 torr (286 cm H₂O) have been documented.³⁴ Airway pressures well above 100 cm H₂O are routinely achieved and held by trumpet players³⁵ without causing barotrauma. Alveolar rupture is extremely rare in wind instrument players, as it is in glass blowers, weight lifters, and other healthy people after coughing, sneezing, or straining.^{3,36}

In addition, for any given airway resistance, varying inspiratory flow will also vary the pressure developed as a given volume is delivered to a patient's lungs, but will not affect end-inspiratory alveolar pressure. For any given volume-targeted inspiratory waveform, shortening inspiratory time will increase peak airway pressure but will not affect overall lung distension if the set tidal volume is unchanged; in fact, higher proximal airway pressures will decrease delivered alveolar volume because of increased compressed volume in the tubing. Conversely, increasing inspiratory time will lower peak airway pressure but could even increase alveolar distension over a series of breaths if insufficient time were provided for complete exhalation, resulting in air-trapping and endogenous positive end-expiratory pressure (auto-PEEP).

It should be remembered that peak inspiratory pressure can also be increased as a result of problems with respiratory system compliance such as low pulmonary parenchymal compliance caused by acute respiratory distress syndrome (ARDS) or pulmonary edema, chest wall problems such as circumferential burns or chest wall edema, and abdominal distension. Such situations will be reflected in an elevated plateau pressure. If this value is high relative to the pressure outside the lung parenchyma—a question that can be sorted out using esophageal manometry³⁷—the transpulmonary pressure difference will be high and the patient may be at increased risk for barotrauma. High plateau pressures in the setting of increased extrapulmonary pressure, on the other hand, will be associated with a small transpulmonary pressure difference and a lower risk of alveolar rupture.

The Importance of Ventilator Mode and Other Settings

To date, no studies have definitively established that the particular ventilator mode or other ventilator settings such as tidal volume or PEEP affect the incidence of barotrauma. The lack of effect secondary to tidal volume in conventional volume-control ventilation can best be seen by comparing the incidence of barotrauma in the era before and after the widespread adoption of lung-protective ventilation in patients with ARDS. In six large series published between 1974 and 1983, including data from 2980 mechanically ventilated patients, the incidence of detectable extraalveolar air ranged from 4% to 11%, with most in the range of 5% to 7%.^{28,30,38–41} Although some patients with acute lung injury and ARDS were included in these series, barotrauma was not specifically examined in those conditions. As [Tables 44-4](#) and [44-5](#) demonstrate, the incidence of barotrauma has not declined significantly since lung-protective ventilation became a part of clinical practice, despite a general impression that this complication is now encountered less frequently than in the past.⁴²

Table 44-4: Incidence of Barotrauma in Randomized Trials Comparing Low and High Tidal Volume Strategies in Patients with Acute Respiratory Distress Syndrome

Study (Reference)	Tidal Volume Used in High Tidal Volume Group (N)	Tidal Volume Used in Low Tidal Volume Group (N)	Incidence of Barotrauma in High Tidal Volume Group: N(%)	Incidence of Barotrauma in Low Tidal Volume Group: N(%)
Amato et al ⁴⁴	12 mL/kg (24)	≤6 mL/kg (29)	10 (42) ^b	2 (7)
Brochard et al ⁴⁵	10 to 15 mL/kg ^b (58)	6 to 10 mL/kg (58)	7 (12) ^c	8 (14)
Stewart et al ⁴⁷	10 to 15 mL/kg ^a (60)	≤8 mL/kg (60)	4 (7) ^d	6 (10)
Brower et al ⁴⁶	10 to 12 mL/kg ^a (26)	≤8 mL/kg (26)	1 (3.8) ^e	2 (7.7)
ARMA Study ⁴³	12 mL/kg ^a (432)	<6 mL/kg (429)	48 (11) ^e	43 (10)
Villar et al ⁴⁸	9 to 11 mL/kg ^a (50)	5 to 8 mL/kg (53)	4 (8.4) ^f	2 (4)
Total	(650)	(655)	74 (11.4)	64 (9.6)

Tidal volume based on actual weight minus estimated weight gain secondary to water and salt retention.

^aTidal volume based on predicted body weight.

Definitions of barotrauma used in included studies:

^bPneumothorax, pneumomediastinum, and subcutaneous emphysema.

^cPneumothorax requiring a chest tube.

^dPneumothorax, pneumomediastinum, pneumoperitoneum, or pneumopericardium on chest radiograph.

^ePneumothorax, pneumomediastinum, pulmonary interstitial emphysema, or pneumatocele ≥2 cm within 28 days of randomization.

^fNot defined.

Table 44-5: Incidence of Barotrauma in Multicenter Randomized Trials in Which Both Groups of Patients Received Lung-Protective Ventilation

Study (Reference)	Control Group (N)	Intervention Group (N)	Incidence of Barotrauma in Control Group: N(%)	Incidence of Barotrauma in Intervention Group: N(%)
Brower et al ⁴⁹	Low PEEP (273)	High PEEP (276)	27 (10) ^b	30 (11)
Mancebo et al ^{50,a}	Supine ventilation (60)	Prone ventilation (76)	4 (6.7) ^c	7 (9.2)
Meade et al ⁵¹	Conventional PEEP; plateau pressure <30 cm H ₂ O (508)	High PEEP; plateau pressure <40 cm H ₂ O (475)	47 (9.1) ^c	53 (11.2)
Mercat et al ⁵²	PEEP 5 to 9 cm H ₂ O (382)	PEEP set to reach plateau pressure 28 to 30 cm H ₂ O (385)	22 (5.8) ^d	26 (6.8)
Papazian et al ⁵³	No paralysis (162)	48 hours of paralysis with <i>cis</i> -atracurium (177)	19 (11.7) ^e	9 (5.1)
Total	(1385)	(1389)	119 (8.6)	125 (9.0)

Abbreviation: PEEP, positive end-expiratory pressure.

All trials used predicted body weight to determine the tidal volume.

^aPatients in this trial were allowed to receive up to 10 mL/kg in the control (supine) and intervention (prone) arms. Average tidal volumes delivered were 8.6 ± 1.6 mL/kg in the supine group and 8.3 ± 1.7 mL/kg in the prone group.

Definitions of barotrauma used in included studies:

^bPneumothorax, pneumomediastinum, pulmonary interstitial emphysema, or pneumatocele ≥2 cm within 28 days of randomization.

^cNew pneumothorax, pneumomediastinum, pneumoperitoneum, or subcutaneous emphysema on chest radiograph or chest tube insertion for known or suspected spontaneous pneumothorax.

^dPneumothorax, defined as needing a chest tube between days 1 and 28 of the study.

^ePneumothorax, pneumomediastinum, subcutaneous emphysema, or pneumatocele ≥2 cm within 28 days of randomization.

In trials comparing low and high tidal-volume strategies,^{43–48} as well as trials in which both arms were managed with lower tidal volumes,^{49–53} the incidence of barotrauma does not differ significantly from that documented in earlier

periods. Further support for a lack of a role for tidal volume comes from studies that examined risk factors for the incidence of barotrauma, which revealed that tidal volume had no significant effect.^{47,54,55}

It is not clear whether other, less commonly used modes of ventilation affect the incidence of barotrauma. Several studies on the use of high-frequency oscillatory ventilation in acute respiratory failure, for example,^{56–58} have documented incidence rates of pneumothorax or barotrauma of 8% to 9%, although a retrospective review of high-frequency oscillatory ventilation use at three centers in Toronto documented a much higher incidence rate of 21.8%.⁵⁹ A randomized trial comparing airway pressure release ventilation to conventional volume-targeted lung-protective ventilation, in patients with acute respiratory failure following trauma, found no difference in the incidence of barotrauma with the two approaches.⁶⁰

The use of noninvasive ventilation has increased substantially in recent years, particularly for the management of acute exacerbations of chronic obstructive pulmonary disease. One report suggests the incidence of barotrauma is less than 5% with this modality,⁶¹ but, overall, less data are available on this question compared to the other support modalities.

By increasing alveolar distension, increased levels of PEEP might be expected to cause more barotrauma. As with tidal volume, however, no studies have established a strong relationship in this regard. Eisner et al⁶² examined 718 patients enrolled in the ARDS Network clinical trials and found that higher levels of PEEP correlated with increased likelihood of developing clinical barotrauma during the first 4 days (relative hazard: 1.5), although the 95% confidence interval included 1 (0.98 to 2.3). More recently, two large randomized studies^{49,51} comparing the use of high and low PEEP strategies in patients with ARDS ventilated at low tidal volumes (≤ 6 mL/kg predicted body weight) showed no significant differences in the incidence of barotrauma. Even if data did suggest an association between PEEP and increased risk of barotrauma, this finding could be interpreted in two ways: either higher levels of PEEP predispose to barotrauma, or sicker patients (who require higher levels of PEEP) are more likely to develop barotrauma.

End-inspiratory plateau pressure is another variable that could affect alveolar distension and, therefore, the risk of barotrauma, but consistent evidence of such a link is also lacking. Boussarsar et al⁵⁴ reviewed the findings of eleven studies (2270 patients) that reported the incidence of barotrauma in patients with ARDS and noted that end-inspiratory plateau pressure was the only ventilator management-related variable that correlated statistically with the occurrence of barotrauma. In an international study of 5183 mechanically ventilated adult patients with a wide variety of diagnoses, however, Anzueto et al⁶³ found no correlation between any ventilator setting or pressure measurement and the development of barotrauma; several recent large randomized trials of ventilator management in ARDS in which plateau pressures were different between the two arms of the study showed no differences in barotrauma incidence (see [Table 44-5](#)).^{43,51,52} One reason why plateau pressure may not be convincingly related to the risk of barotrauma may be because plateau pressure represents the difference between pressure in the airways and the atmosphere and, hence, total respiratory system compliance, whereas the risk of barotrauma is more a function of the pressure difference between the alveoli and bronchovascular sheath. Plateau pressure may be high but if this is caused by factors outside the lungs, such as increased abdominal distension or chest wall edema, the elevated pressures may not lead to alveolar rupture.

The Importance of Proper Manual (Bag) Ventilation

A circumstance in which tidal volumes and airway pressures are delivered substantially above desirable levels is manual (bag) ventilation, an adjunctive ventilation method used immediately following intubation, during cardiopulmonary resuscitation, in association with airway suctioning, and chest physiotherapy, and during patient transport.⁶⁴ The

volumes delivered and the pressures generated are typically not monitored and can vary substantially, depending on the size of the operator's hands, whether one hand or two is used, and other factors. Tidal volumes during manual ventilation ranged between 838 mL and 1674 mL in one study, with a mean value of 170% of the ventilator's set tidal volume.⁶⁵ In a bench study, in which experienced respiratory therapists performed manual ventilation on a lung model to simulate ventilating a 70-kg patient, delivered tidal volumes varied from 400 mL to more than 1000 mL, with airway pressures sometimes exceeding 100 cm H₂O.⁶⁶ Indeed, the development of clinical barotrauma during manual ventilation has been reported several times.^{67–69}

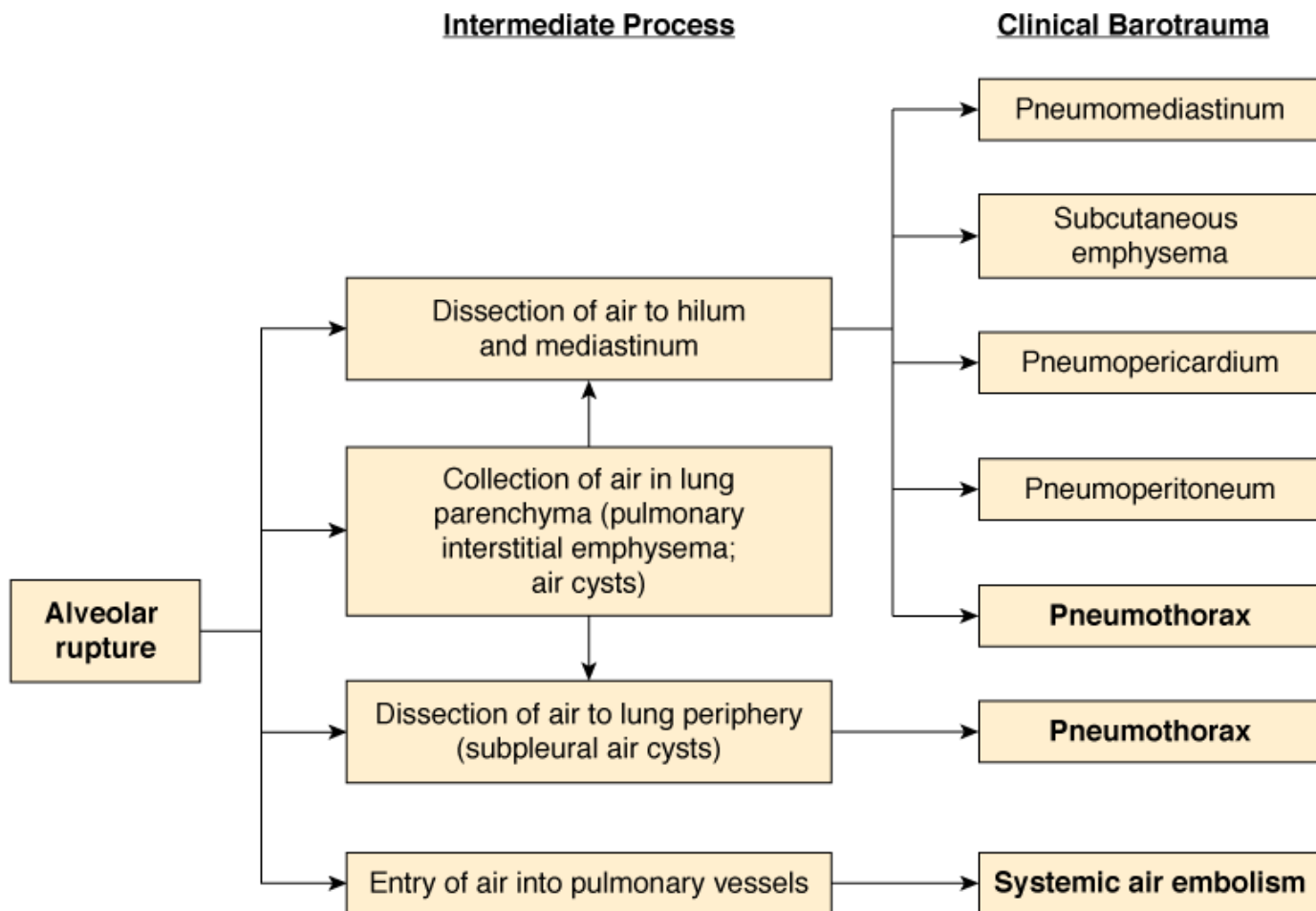
The Importance of the Underlying Disease Process

The occurrence of alveolar rupture during mechanical ventilation is influenced markedly by the presence and nature of underlying lung pathology. Barotrauma is rare in patients with normal lungs—for example, in routine postoperative ventilation³⁹ or in paralytic states such as high cervical spinal cord injury or Guillain-Barré syndrome. In contrast, barotrauma is more common in patients with underlying obstructive or restrictive lung disease.^{42,63} In an observational cohort study of 5183 patients who underwent mechanical ventilation for more than 12 hours in 361 ICUs in twenty countries,⁷⁰ barotrauma was observed in 154 patients (2.9%).⁶³ The incidence of barotrauma, defined as pulmonary interstitial emphysema, pneumothorax, pneumomediastinum, pneumoperitoneum, or subcutaneous emphysema, was 2.9% in patients with chronic obstructive pulmonary disease, 4.2% in pneumonia, 6.3% in asthma, 6.5% in ARDS, and 10.0% in chronic interstitial lung disease.⁶³ Logistic regression analysis identified the last three of these conditions as independent risk factors for development of barotrauma. An earlier analysis by Gammon et al⁷¹ of 168 consecutive patients who underwent mechanical ventilation at one center found with multivariate analysis that of all variables examined, only the presence of ARDS correlated with development of pneumothorax.

The Spread of Extraalveolar Air Once Alveolar Rupture Occurs

Because pressures in the interstitium and bronchovascular sheath tend to be slightly less than those in the alveoli at peak lung inflation, there is a tendency for air to move into these areas once the alveolar walls are disrupted.^{4–6} Although this is also accepted as the mechanism of the spread of extraalveolar air in spontaneously breathing patients, it is particularly facilitated by positive-pressure ventilation, especially when large tidal volumes are used. Figure 44-2 depicts the possible routes of spread once air leaves the alveolus. Except for systemic air embolism, which is uncommon, and pulmonary interstitial emphysema, the common pathway for all the clinical forms of barotrauma is pneumomediastinum.⁸ Once in the mediastinum, air follows the path of least resistance and may rupture through the delicate mediastinal fascia and overlying pleura into the pleural space. Why this occurs in some patients and not in others may be determined by local pleural scarring in some cases, but is usually not readily apparent.

Figure 44-2



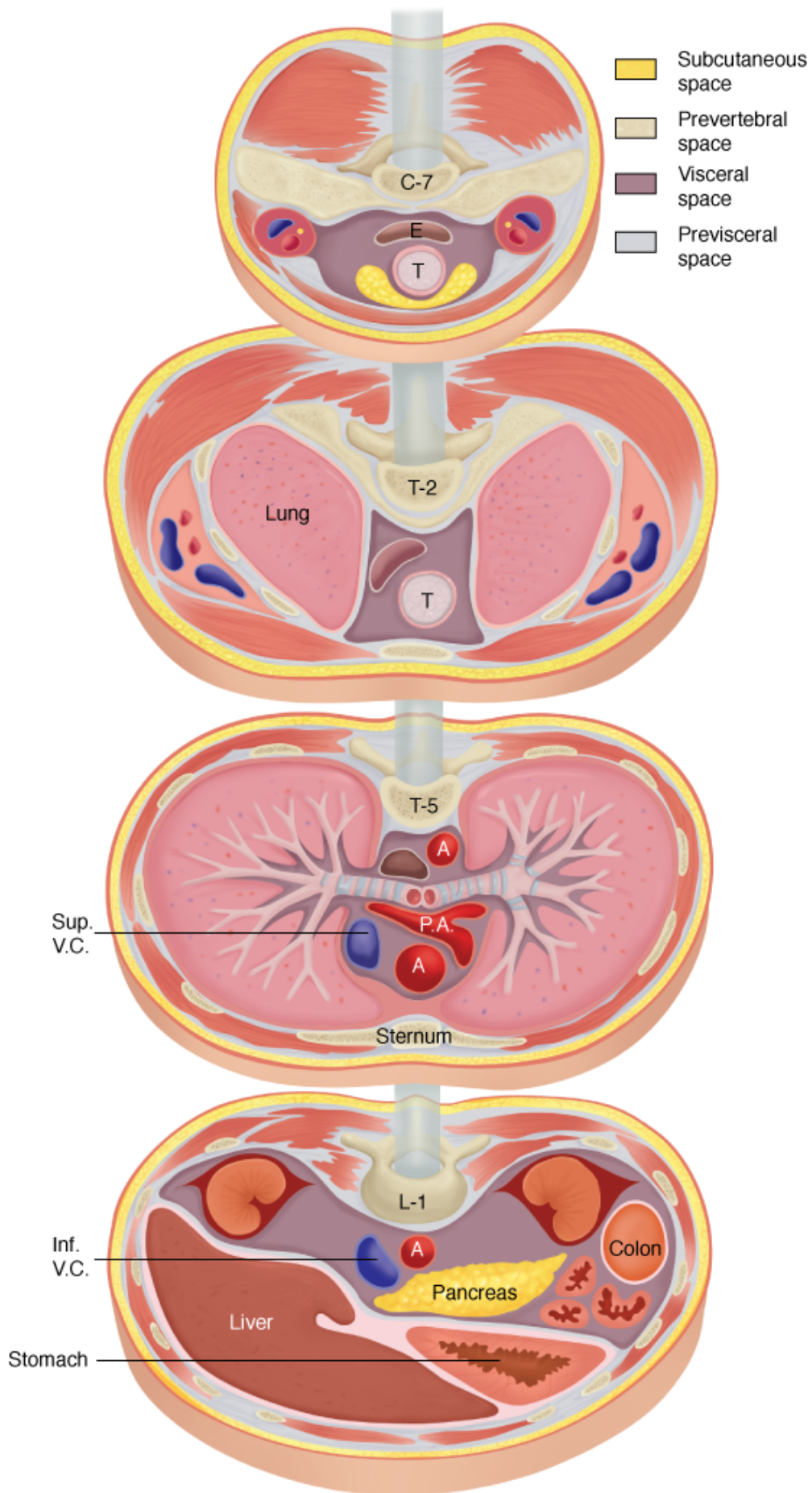
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Pathogenesis of the various forms of barotrauma, with those of greatest clinical importance in bold type. In addition to being associated with use of mechanical ventilation, pneumothorax can also occur as a result of puncture or laceration of the visceral pleura during central line placement or other procedures or traumatic injuries such as rib fractures.

The spread of extraalveolar air to a wide variety of locations in the body can be understood through reference to the fascial planes of the neck, mediastinum, and retroperitoneum, as shown in [Figure 44-3](#).⁴ Three distinct cervical compartments exist—a previsceral space lying between the deep cervical fascia and the pretracheal fascia, a visceral space lying between the pretracheal fascia and the prevertebral fascia, and a prevertebral space situated behind the prevertebral fascia. These cervical compartments continue via the mediastinum through the thorax and trunk, providing the potential for the spread of air into the retroperitoneal space.

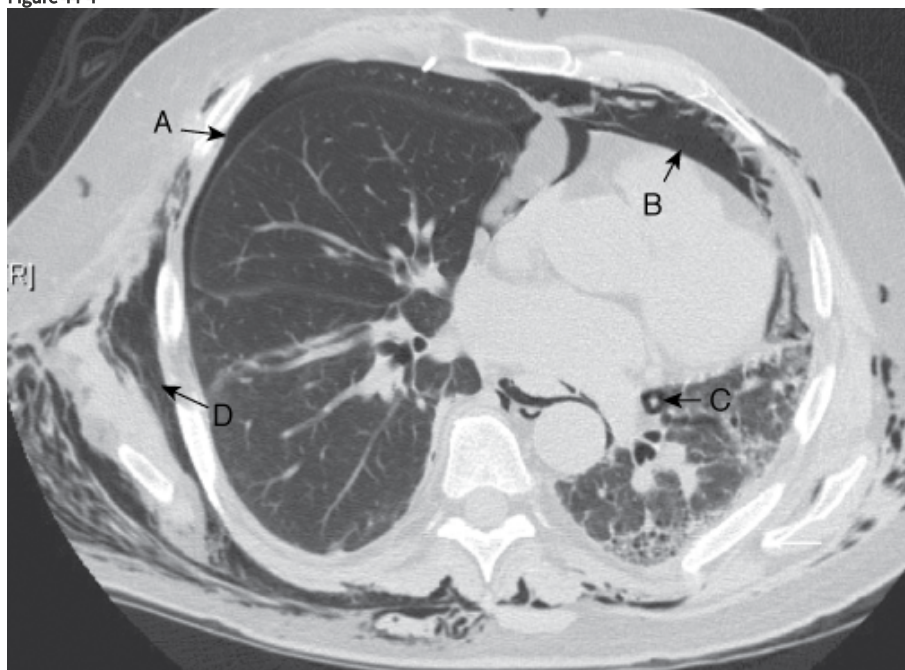
Figure 44-3



Fascial planes of the neck, mediastinum, and retroperitoneum, as depicted at the levels of C-7, T-2, T-5, and L-1, respectively. Because the soft-tissue compartments are continuous, air can dissect into a wide variety of anatomic locations. (Adapted, with permission, from Maunder et al.⁴) Abbreviations: A, aorta; E, esophagus; inf.V.C., inferior vena cava; P.A., pulmonary artery; Sup.V.C., superior vena cava; T, trachea.

Air may continue to leave the airways, as shown in [Figure 44-1](#), so long as the alveolar rent is large enough and a sufficient pressure gradient exists. If this path of least resistance leads up the mediastinum and into the soft tissues of the neck and upper chest, subcutaneous emphysema may become massive, spreading extensively over the body. Occasionally the pressure gradient and local anatomy favor passage of air into the retroperitoneal space, from which it decompresses into the peritoneum,⁷²⁻⁷⁶ causing massive abdominal distension. When the extraalveolar air has reached the pleural cavity, the resultant pneumothorax may rapidly increase with successive positive-pressure breaths. Air readily leaves the torn alveolus and passes into the pleural space, but cannot return between inspirations because of the collapsibility of the lung and mediastinum. A ball-valve mechanism is created and fatal tension pneumothorax can rapidly result. [Figure 44-4](#) depicts examples of the various forms of barotrauma that can result from these mechanisms.

Figure 44-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Computed tomography image through midchest showing multiple manifestations of pulmonary barotrauma in the same patient including: (A) pneumothorax; (B) pneumomediastinum; (C) pulmonary interstitial emphysema represented by a halo of air surrounding a vascular structure; and (D) subcutaneous emphysema.

Bronchopleural Fistula

After evacuation of a pneumothorax via tube thoracostomy, air may continue to leak into the pleural space and via external suction into the pleural collection device. Such a bronchopleural air leak tends to be perpetuated by the very measures often required to reinflate the collapsed lung and to maintain gas exchange in patients with severe acute respiratory failure. Once a rent exists at either the alveolar or visceral pleural level, the higher the pressure gradient between airways and pleural space, the greater will be the tendency of the bronchopleural fistula to leak. It therefore stands to reason that management should focus on decreasing airway pressure and minimizing pleural suction. As

discussed later (see the section [Management: Bronchopleural Fistula](#)), however, little experimental evidence is at hand to support this hypothesis, and the management needs of critically ill patients often prevent the realization of these goals.

Clinical Manifestations

The different clinical forms of barotrauma shown in the right-hand column in [Figure 44-2](#) vary considerably in their clinical manifestations, relative frequency, and potential for doing harm to the patient. [Table 44-6](#) lists the nine categories of extraalveolar air that are discussed in the remainder of this chapter, and shows their relative frequency in ventilated patients, along with their relative seriousness and potential threat to life. With few data from prospective studies of the incidence of the different forms of barotrauma across the broad range of patients managed on mechanical ventilation, the frequencies listed in [Table 44-6](#) are based on our clinical experience and what little information can be gleaned from the literature. Tension pneumothorax and systemic air embolism are always potentially fatal events. Beyond these, however, the effects of barotrauma on survival, organ dysfunction, ICU stay, and other outcomes, separate from those of the underlying disease, are largely unknown.

Table 44-6: Clinical Forms of Barotrauma during Mechanical Ventilation

Form	Relative Frequency	Potential Threat to Life
Pulmonary interstitial emphysema	++++	+
Intraparenchymal air cysts	++	++
Pneumomediastinum	++++	++
Pneumopericardium	+	+++
Subcutaneous emphysema	+++	+
Pneumoperitoneum	+	++
Systemic air embolism	+	++++
Pneumothorax	+++	++++
Bronchopleural fistula	++	++

Pulmonary Interstitial Emphysema and Cystic Dilation

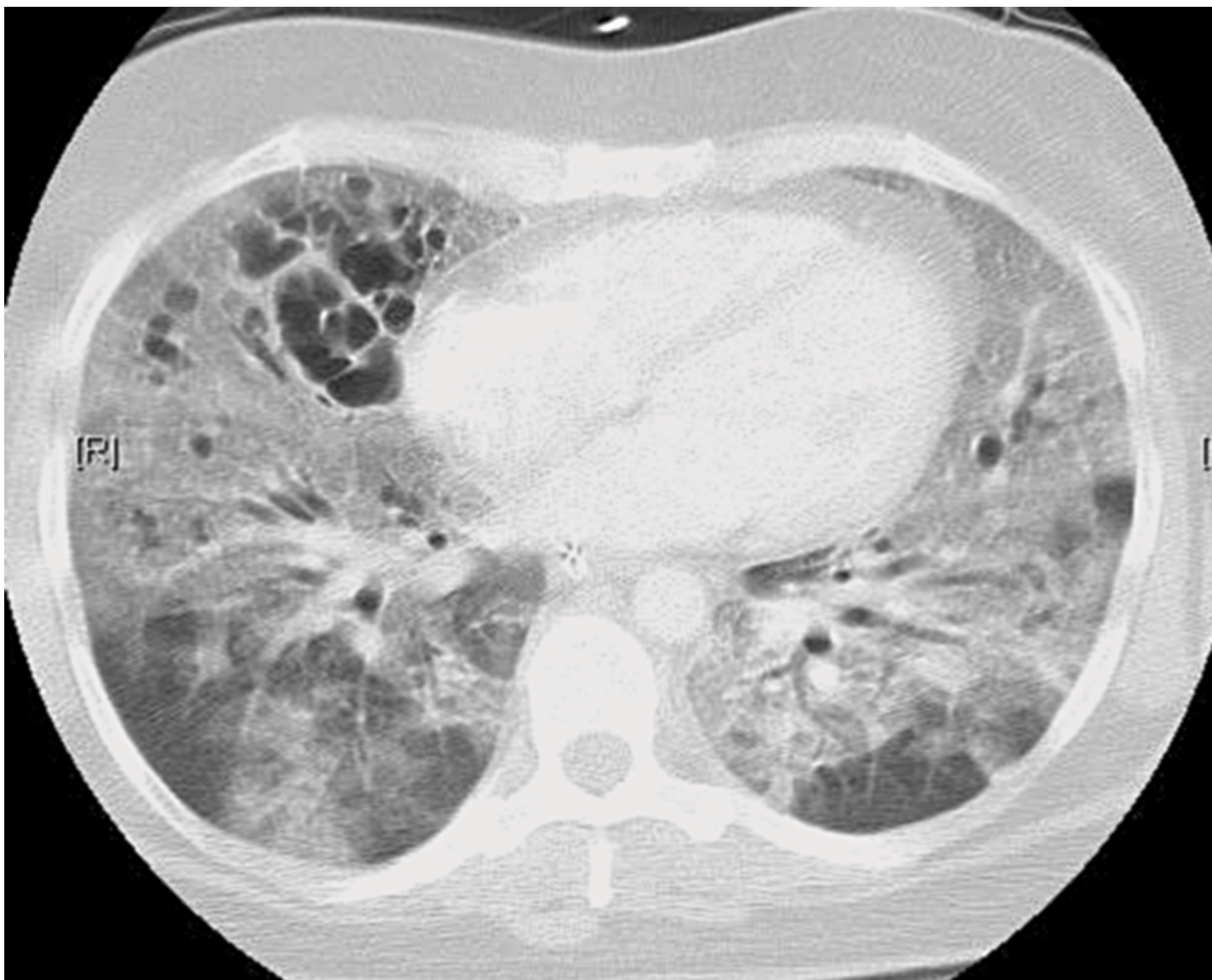
Air in the interstitium of the lung can be identified at autopsy in patients who die with clinical barotrauma associated with positive-pressure ventilation, and is easily demonstrated using computed tomography (CT) scanning (see [C](#) in [Fig. 44-4](#)).⁷⁷⁻⁷⁹ With appropriate technique pulmonary interstitial emphysema may also sometimes be detected on plain radiographs of the chest, and be present without other signs of extraalveolar air.^{80,81} A variety of patterns have been described.⁸¹ Parenchymal stippling, thought to represent multiple small pulmonary vessels and their air-distended

vascular sheaths in cross section, has been referred to as a “salt-and-pepper” pattern.⁸⁰ Air dissection within the pulmonary interstitium is believed responsible for lucent mottling,⁸¹ with small cyst-like lucencies that may increase progressively in size on successive films. Other signs of pulmonary interstitial emphysema include lucent streaks, believed to be produced by air dissecting toward the hilum along the course of larger pulmonary vessels, and perivascular halos in the perihilar regions of the lung, representing such air collections seen end-on. All these signs are most readily seen in the presence of diffuse pulmonary consolidation or atelectasis, which increases the radiographic contrast between parenchyma and extraalveolar air.

Although it is well-defined pathophysiologically (see [Figs. 44-1](#) and [44-2](#)), pulmonary interstitial emphysema in mechanically ventilated adult patients has doubtful practical clinical importance, primarily because of the realities of bedside imaging in the ICU. Several older studies described pulmonary interstitial emphysema in critically ill patients,^{30,31,82-84} and in some instances it has been shown to precede the development of pneumothorax. The subtle changes identifying this entity, however, are beyond the resolution of most bedside radiologic tools. In one study attempting to elucidate the patterns and risk factors for barotrauma in ventilated patients, the investigators abandoned their initial plan to determine whether pulmonary interstitial emphysema precedes pneumothorax because they could not reliably or reproducibly detect the former.⁸⁵ In our experience, the degree of resolution and interexposure consistency required to detect and follow this radiographic finding are beyond the capabilities of most portable X-ray units, that must frequently shoot through bandages and bedding, with suboptimal positioning and patients who cannot hold still or take a deep breath.

Intrapulmonary cystlike air collections in patients with ARDS have been recognized with increased frequency since the use of CT in such patients became commonplace. The early studies of Maunder et al⁸⁶ and Gattinoni et al⁸⁷ demonstrated the heterogeneous nature of lung involvement in ARDS, and especially late in the course of this disorder large collections of free air within the pulmonary parenchyma are common ([Fig. 44-5](#)). Pneumothorax also seems to be common in this setting, although no study published to date has documented this observation objectively.

Figure 44-5



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CT image through the midchest in a patient with ARDS receiving mechanical ventilation with PEEP. Parenchymal involvement is markedly inhomogeneous, with relatively normal areas in the dependent lung zones, widespread ground glass opacities in both lungs and intraparenchymal cystic air collections, primarily in the anterior portions of the right lung.

Small subpleural air collections, also identified on CT scans occur frequently during the course of ARDS.⁸⁸ These collections probably represent centrifugal dissection of air through the interstitium to the visceral pleura, and are the likely source for pleural air in some patients.

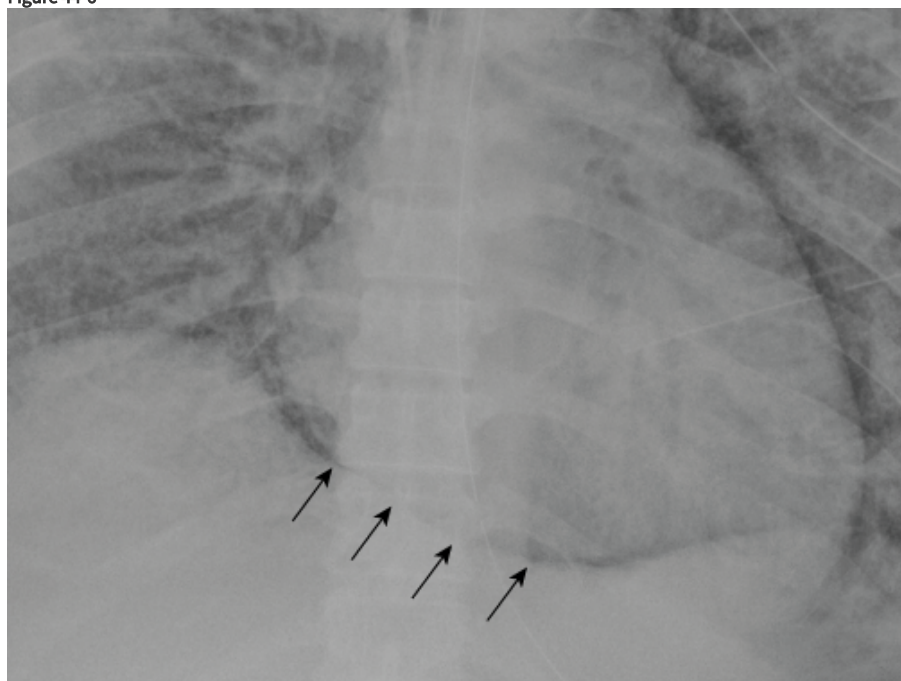
Pneumomediastinum

The mechanisms for alveolar rupture and the spread of air to the mediastinum illustrated in [Figures 44-1](#) and [44-2](#) apply to spontaneously breathing, nonintubated patients as well as to those receiving mechanical ventilation.⁸ In the former, symptoms referable to pneumomediastinum are frequent and characteristic. Stabbing precordial chest pain is the most common complaint,⁸⁹⁻⁹¹ occurring in 80% to 90% of patients.^{4,92} “Hamman’s crunch,” a crunching or clicking sound best heard over the retrosternal area synchronous with cardiac systole,⁹³ can be detected in many patients who present

with spontaneous pneumomediastinum. It is not specific, however, and may occur in patients with spontaneous pneumothorax without evidence of pneumomediastinum.⁹⁴

Symptoms may be unobtainable from an intubated, critically ill patient, however, and the physical findings characteristic of pneumomediastinum may be hard to detect amid the hubbub of the ICU. Most often, pneumomediastinum is first detected radiologically, usually as an incidental finding. The most common appearance is a radiolucent area paralleling the left-heart border. This is readily detected in the presence of fluid density in the adjacent lung tissue, as with pneumonia or pulmonary edema. A thin, vertically oriented white line representing the mediastinal pleura may be seen when the pneumomediastinum is bordered by normal lung parenchyma. Once detected, a pneumomediastinum can often be traced cephalad beyond the hilum and into the neck. Other radiographic findings include highlighting of the aortic knob, which is surrounded by more lucent gas density than usual, and the “continuous diaphragm sign” described by Levin (Fig. 44-6).⁹⁵ Many patients with ventilator-associated pneumomediastinum also develop subcutaneous emphysema, which is typically more obvious radiographically.

Figure 44-6



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The “continuous diaphragm sign” of pneumomediastinum in a patient with barotrauma complicating ARDS. An unbroken radiolucent line denoted by the *arrows* extends from one hemidiaphragm to the other, rendering the inferior heart border clearly visible.

Pneumomediastinum is frequently the first sign of barotrauma, and a substantial number of cases go on to development of pneumothorax. In one retrospective study,⁸⁵ thirty-four of 139 intubated patients had radiographic signs of barotrauma at some point in their ICU course. Pneumomediastinum was the initial manifestation in twenty-four of the thirty-four patients, and a pneumothorax was subsequently diagnosed in ten (42%).

Pneumopericardium

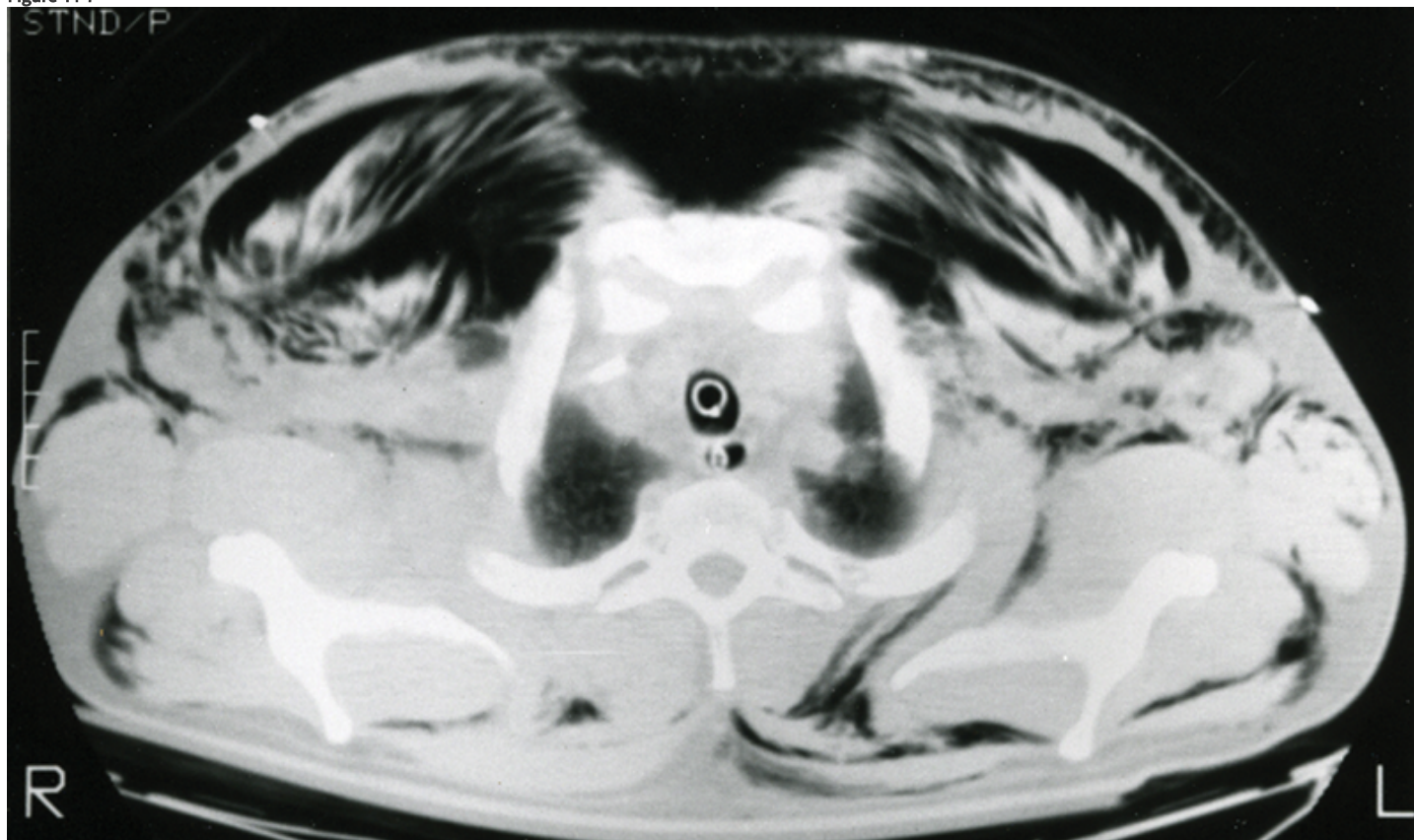
Dissection of air into the pericardium can follow alveolar rupture, and pneumopericardium may occasionally be the only sign of barotrauma. Long known as a complication of respiratory distress syndrome in neonates, it has been reported in

adults as well,^{96,97} particularly following blunt chest trauma,^{98,99} and complicating severe bacterial pneumonia.^{100,101} Although hemodynamically significant pneumopericardium in ventilated adults is not generally considered likely, presumably because the adult pericardium communicates less readily with the rest of the mediastinum than is the case in neonates, one group⁹⁶ reviewed eighty-one previously published cases and found 37% of them to have been hemodynamically significant. Several patients in this series were said to have required emergency pericardiocentesis. Adding to the uncertainty about the frequency and clinical importance of pneumopericardium in ventilated adult patients is the difficulty often encountered by portable radiographic devices to distinguish air in the pericardium from the much more common finding of pneumomediastinum.

Subcutaneous Emphysema

Although subcutaneous emphysema is commonly observed both radiographically and on physical examination around chest tube insertion sites, most often this form of barotrauma is encountered in the neck and upper anterior chest as it is vented from the superior mediastinum. When present in only small amounts, subcutaneous air produces the typical findings of crepitance on palpation. If the leak continues, as may be seen in severe ARDS when excessive tidal volumes and high inflation pressures are used, obvious distortion of the neck and anterior chest wall can be seen, and palpation shows a characteristic “pneumatic sponginess” as the skin and subcutaneous tissues are depressed to touch the underlying ribs. Air is commonly noted outlining the pectoralis musculature on routine chest radiology; chest CT reveals the spread of air throughout the soft tissues and in all fascial planes (Fig. 44-7).

Figure 44-7

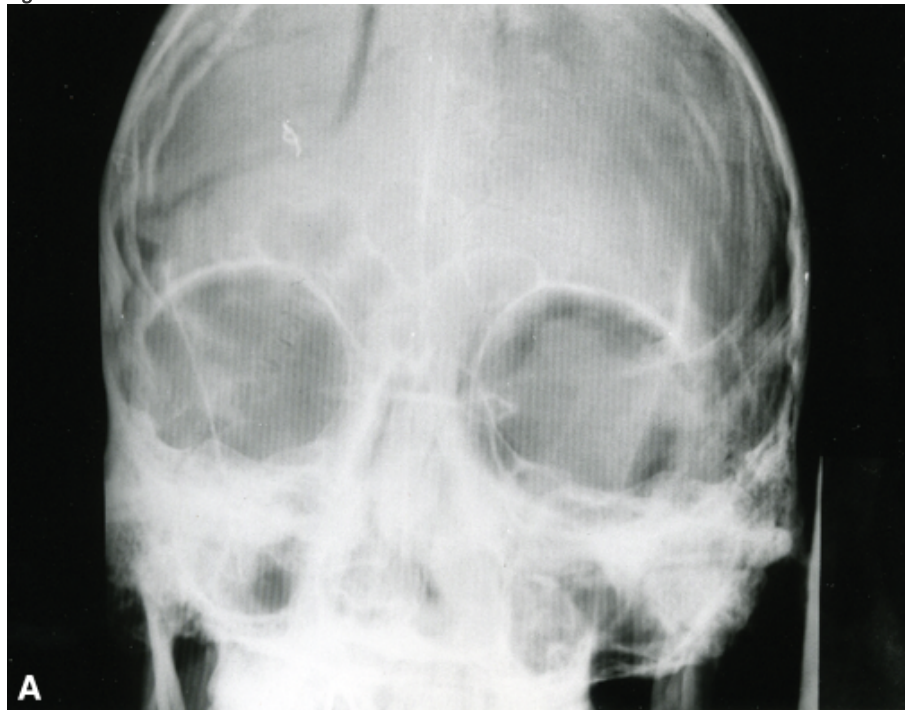


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Example of widespread subcutaneous emphysema in a patient with barotrauma complicating ventilator management in severe ARDS. CT image through the shoulders, at the most cephalad extension of the lung apices, showing air widely dispersed under the skin and outlining the musculature.

From the neck and chest wall the air can spread literally everywhere in the body, collecting especially in areas with loose subcutaneous tissue, such as the upper arms and axillae and the abdominal wall. Periorbital accumulation may become extensive enough to force the eyelids shut. Radiographic studies obtained for other reasons in patients with widespread subcutaneous emphysema may reveal air in a variety of unexpected locations (Fig. 44-8).

Figure 44-8



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Examples of widespread subcutaneous emphysema in patients with barotrauma complicating ventilator management in severe ARDS. **A.** A bedside skull film showing air beneath the scalp on the left and in the orbit surrounding the ocular

globe. A craniotomy defect can be seen on the right. **B.** Demonstration of tension pneumoscutum in a different patient with widespread extraalveolar air including pneumoperitoneum and extensive subcutaneous emphysema.

Although it may be uncomfortable for the patient and distressing to the caregivers at the bedside, subcutaneous emphysema is nearly always clinically benign. Air in the subcutaneous tissues or dissecting along deeper fascial planes poses little if any direct threat to the adjacent tissues. The air tends to collect in areas with loose connecting tissue, whose blood supply is not compromised by the stretching and other distortion. Like pneumomediastinum, this form of barotrauma serves mainly as an indicator rather than constituting a management problem per se. It signifies the presence of a substantial air leak, and should alert the clinician to the possibility of pneumothorax, but therapy for the subcutaneous air itself is seldom indicated. Airway compromise¹⁰² hemodynamic collapse,¹⁰³ and manifestations of abdominal compartment syndrome,¹⁷ have been reported in the context of large-airway disruptions and the use of high inflation pressures, although these instances are rare.

Pneumoperitoneum

Figure 44-3 shows how air from the lung that reaches the mediastinum can spread to the retroperitoneal space and thence into the peritoneum itself. The appearance of free air in the peritoneal cavity in a patient receiving mechanical ventilation⁷⁴ poses a diagnostic dilemma, even when subcutaneous emphysema and other signs of the wide spread of extraalveolar air are present. Avoidance of an exploratory laparotomy or other invasive procedure may be exceedingly difficult in such circumstances. Determining the partial pressure of oxygen (P_{O_2}) of aspirated peritoneal gas has been reported as a means of differentiating pneumoperitoneum from a ruptured abdominal viscus from air dissecting into the peritoneum under positive-pressure ventilation after alveolar rupture,¹⁰⁴ although the accuracy of this procedure has not been validated by others.

Although rare, accumulation of air in the peritoneum as a result of barotrauma during positive-pressure ventilation can cause or contribute to increased intraabdominal pressure⁷² and, conceivably, abdominal compartment syndrome. Generally, however, air in the retroperitoneal and peritoneal spaces poses no intrinsic threat to health, and produces no physical signs other than subcutaneous emphysema and abdominal distension. In exceptional instances, air from ventilator-associated barotrauma can even dissect into the scrotum (see Fig. 44-8B) presumably via the peritoneum.

Systemic Air Embolism

Systemic air embolism may occur more commonly as a manifestation of barotrauma during mechanical ventilation than has previously been believed. Marini and Culver¹⁰⁵ describe the cases of two young adult patients with ARDS, necrotizing pneumonia, and prior pneumothoraces, in whom enlarging intraparenchymal air cysts and pulmonary interstitial emphysema were noted. While in a semi-upright position, each of these patients developed sudden neurologic events (facial twitching, hemiparesis, and seizures), evidence of acute myocardial injury, and focal livedo reticularis over the shoulder and anterior chest. Focal angioedema of the face subsequently developed. No other explanation for these events could be found, and in each patient there was evidence for recurrent episodes of air embolism. One patient survived without long-term sequelae; the other died of refractory hypoxemia and hypotension. Other cases of fatal air embolism as a manifestation of barotrauma have been reported.^{106,107} In one instance, a patient being ventilated with PEEP for ARDS was found at postmortem examination to have air in the coronary and cerebral circulations, as well as in the pulmonary veins.¹⁰⁷

Morris et al¹⁰⁸ reported the occurrence of venous air embolism via the inferior vena cava, as detected with transesophageal echocardiography, in three surgical patients, two with ARDS and one with multiple trauma. One 26-year-old patient developed an acute myocardial infarction on the thirteenth day of ARDS, while on 30 cm H₂O of PEEP, and transesophageal echocardiography revealed continuous bubbling of air in the inferior vena cava. In the other two patients transesophageal echocardiography revealed inferior vena cava bubbles without prior clinical signs of systemic air embolism. Each of these patients had had previous evidence for extraalveolar air, and the authors postulated that the bubbling originated from subcutaneous or intraperitoneal air that gained entry into the splanchnic venous system.

Several cases of cerebral air embolism have been seen in association with mechanical ventilation in patients with traumatic lung contusion.^{109,110} In several reports, gas bubbling has been observed in the left atrium and left ventricle (but not on the right side) during ventilation with large tidal volumes and PEEP, with cessation during apnea and disappearance or diminution when lower tidal volumes and PEEP were used.^{110,111} Cerebral air embolism has also been reported as a feature of barotrauma in patients receiving noninvasive ventilation via face mask.^{112,113}

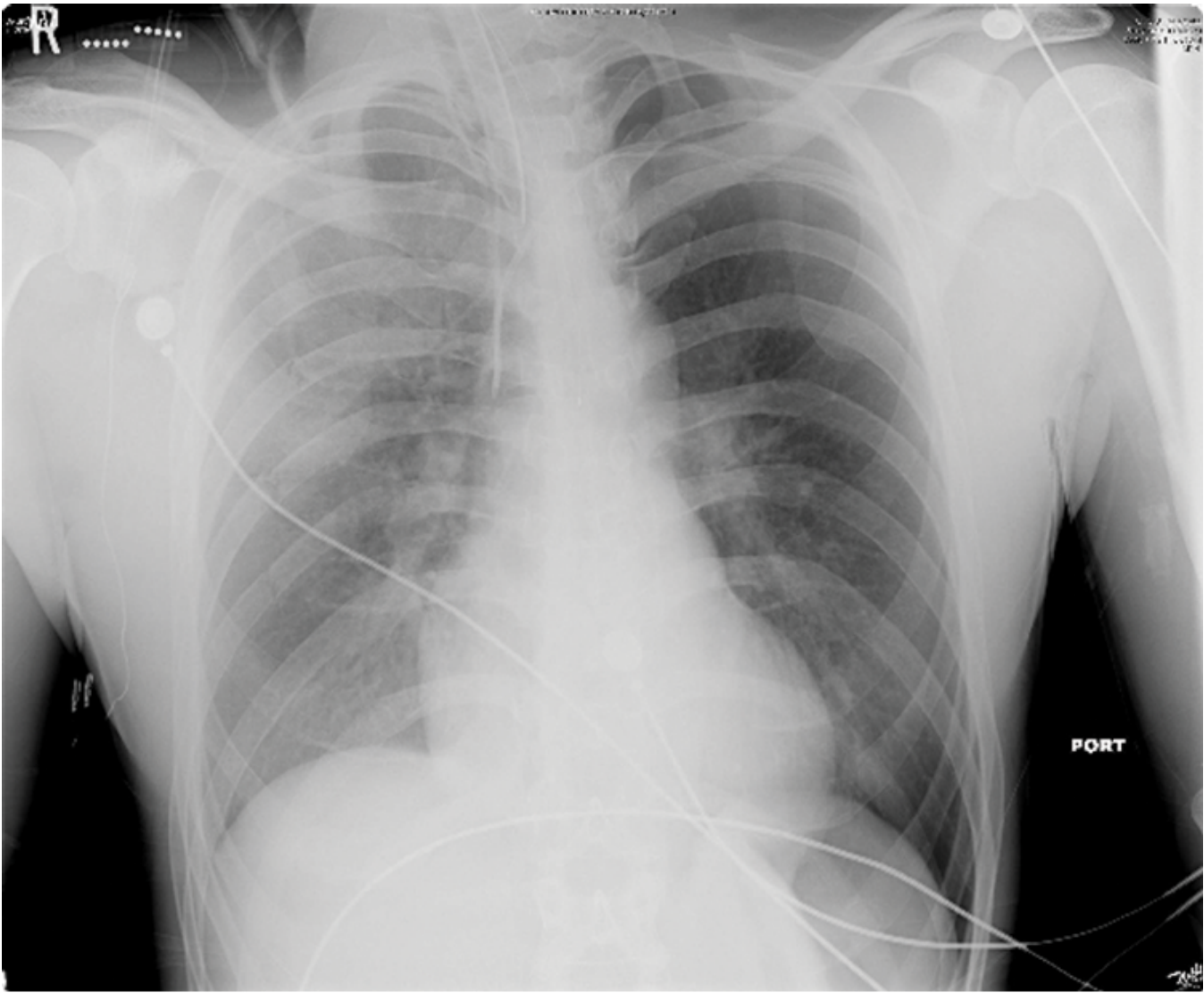
In some patients, once alveolar rupture occurs during mechanical ventilation, decompression into the bronchovascular sheath by the mechanism described by Macklin^{5,6} may fail to occur, allowing pulmonary interstitial air to accumulate under pressure as tension air cysts.⁸⁸ Development of such cysts has been observed to precede the appearance of tension pneumothorax in several instances.^{88,114} In ARDS, Marini and Culver¹⁰⁵ postulate that, in the presence of such intraparenchymal air collections, the strong retractive tendency of the infiltrated parenchyma may tether open vascular channels that have been disrupted by inflammation or shear stress, allowing entry of air.

Systemic air embolism should be considered when patients receiving positive-pressure ventilation (especially for ARDS or with high levels of PEEP) develop unexplained agitation, changed mental status, focal neurologic findings, seizures, or disturbances in cardiac rhythm. The finding of focal livedo reticularis, particularly over the neck, shoulder, or anterior chest wall, should raise the strong suspicion of this complication, as should the occurrence of focal angioedema of the face or neck.

Pneumothorax

Pneumothorax in patients receiving positive-pressure ventilation manifests in three ways. Commonly it is first detected on a routine chest radiograph without clinical signs. The findings may be subtle. Typical radiographic signs of pneumothorax, such as a rim of lucency without lung markings completely surrounding the lung, demarcated by the visceral pleura and denser lung tissue, may be absent. Often the only indication of free pleural air is an increase in overall lucency in one hemithorax or one area of the lung, particularly the lower zone (Fig. 44-9). In some cases the deep sulcus sign, (Fig. 44-10) seen when radiographs are taken with the patient in the supine position, is the only clue to the presence of the pneumothorax. Physical signs of pneumothorax (diminished chest wall excursion, increased percussion resonance, and diminished breath sounds) can rarely, if ever, be detected in a patient who does not have a tension pneumothorax.

Figure 44-9

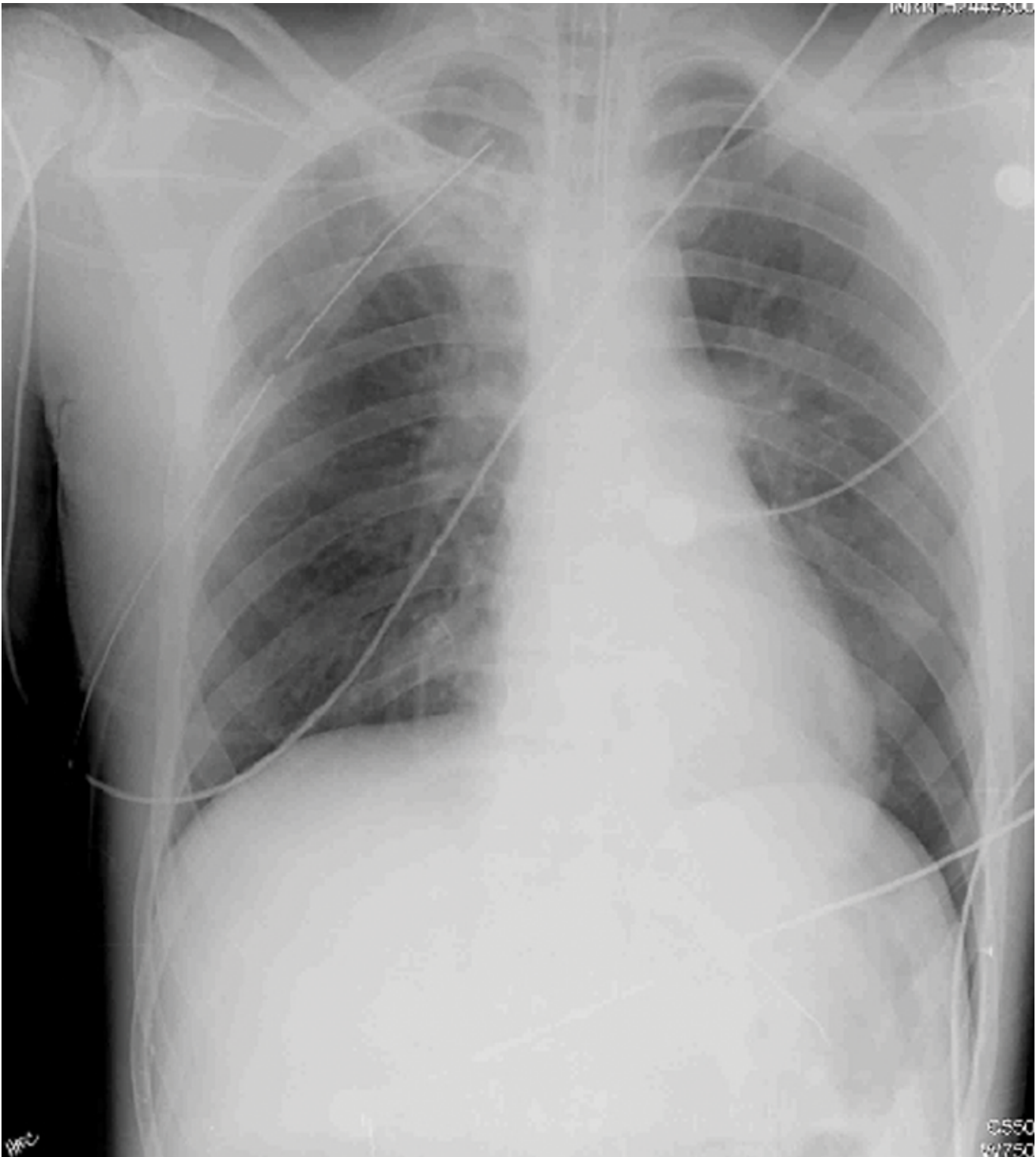


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Loculated left-sided anterior pneumothorax in a patient receiving mechanical ventilation. Lung markings are present throughout the left hemithorax and there is no visible air laterally between the parietal and visceral pleurae. With anterior pneumothorax the only signs may be an increase in overall lucency in one hemithorax, typically confined to the lower lung field, and an increase in the clarity of the left-heart border.

Figure 44-10



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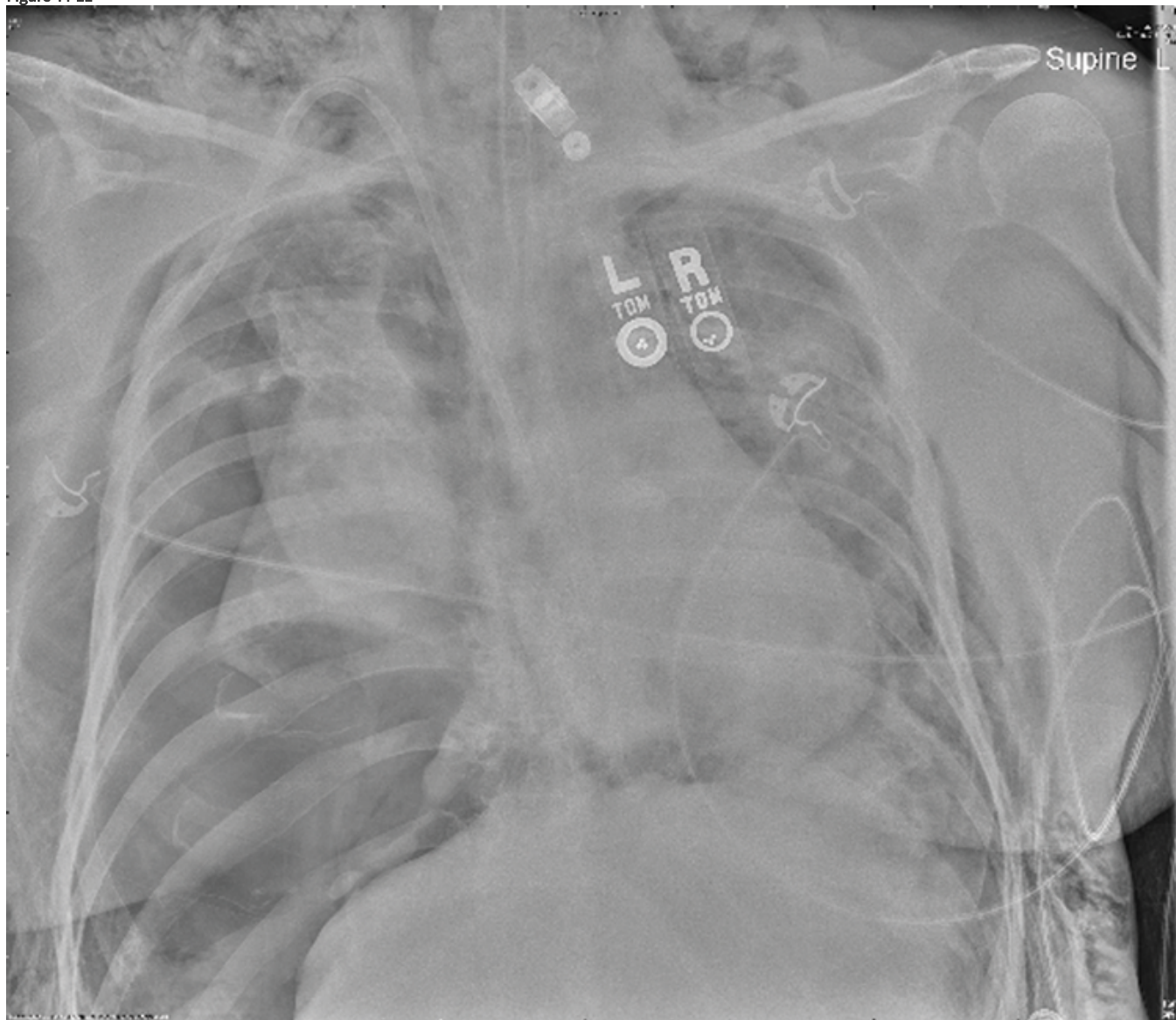
The deep-sulcus sign in a patient receiving mechanical ventilation in the supine position following trauma. The costophrenic sulcus on the left is substantially deeper than the sulcus on the right. There is no visible air laterally between the parietal and visceral pleurae on the left. The patient already has a chest tube on the right side from a pneumothorax detected on that side at the time of admission.

The second presentation for pneumothorax in the ventilated patient is with a change in laboratory or bedside monitoring findings, but without an obvious change in the patient's clinical state. Worsening oxygenation,¹¹⁵ an increase in peak inspiratory airway pressure with accompanying rise in the plateau pressure, or a fall in lung–chest wall compliance may

signal the development of a pneumothorax that has affected pulmonary function but has not yet become hemodynamically significant.

Tension pneumothorax is the third presentation (Fig. 44-11). This is primarily a bedside diagnosis, in that the complication is defined better by its adverse effects on cardiovascular function than by its radiographic characteristics. The latter include mediastinal displacement away from the pneumothorax, reversal of the diaphragmatic curve, and overall enlargement of the affected hemithorax, but these signs can occur in the absence of adverse physiologic effects, and the physiologic effects of pleural tension may be present without them. Most commonly the patient with tension pneumothorax presents with agitation and respiratory distress, hypotension, and other signs of cardiovascular collapse. Although tension pneumothorax is uncommon in spontaneously breathing patients, even in the presence of total lung collapse, the clinician should assume that every radiographically detected pneumothorax in a patient on mechanical ventilation can rapidly accumulate under tension.

Figure 44-11



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Tension pneumothorax in a patient receiving mechanical ventilation for ARDS. There is a large pneumothorax on the right with associated collapse of the right lung and shift of the heart and mediastinal structures toward the left side of the chest. A smaller left pneumothorax is also present. Although features of chest radiographs, including shift of the

mediastinal structures, are strongly suggestive of the diagnosis of tension pneumothorax, the diagnosis is typically made clinically at the bedside before chest radiography is performed.

Tables 44-7 and 44-8 list a number of clinical settings in which acute pneumothorax should be suspected or anticipated in the ventilated patient.

Table 44-7: When to Suspect a Pneumothorax in the Ventilated Patient

Clinical change in patient’s status
Sudden or progressive increase in inspiratory peak or plateau airway pressure
Hypotension or cardiovascular collapse
Sudden onset of agitation and respiratory distress (“fighting the ventilator”)
Suggestive findings on chest radiograph
General increase in volume of one hemithorax
Deep sulcus sign: downward displacement of costophrenic angle and/or hemidiaphragm
Increase in relative radiolucency of one lung or part of one lung

Table 44-8: Clinical Situations Suggestive of a High Risk of Pneumothorax

Use of large tidal volumes (e.g., >8 to 10 mL/kg) in patients with acute lung injury or underlying chronic pulmonary disease
Use of high levels of PEEP (e.g., >15 cm H ₂ O)
High peak airway pressure (e.g., >50 to 60 cm H ₂ O), especially if increasing on the same inspiratory flow settings
Acute respiratory distress syndrome (ARDS), especially late in its course (e.g., 2 to 3 weeks)
Severe underlying obstructive lung disease (COPD, asthma)
Severe bullous lung disease (e.g., idiopathic congenital giant bullous emphysema)
Pulmonary infection complicating ARDS
Pulmonary parenchymal disease accompanied by cystic changes (e.g., <i>Pneumocystis jiroveci</i> pneumonia, lymphangioleiomyomatosis, cystic fibrosis)
Presence of pulmonary interstitial emphysema or pneumomediastinum, and especially their new development on serial chest radiographs, suggesting alveolar rupture and increased likelihood of spread to pleural space
<i>Abbreviations:</i> ARDS, acute respiratory distress syndrome; COPD, chronic obstructive pulmonary disease; PEEP, positive end-expiratory pressure.

Bronchopleural Fistula

The presence of a bronchopleural air leak following insertion of a chest tube could potentially create several important clinical problems (Table 44-9). Unfortunately, relatively little is known about the natural history of this entity, and few data exist to substantiate the individual points in the table.

Table 44-9: Potential Adverse Effects of Bronchopleural Fistula in the Ventilated Patient

Effect	Problems	Comment
Incomplete lung expansion	Atelectasis Worsened ventilation–perfusion mismatching ?Failure of leak to close	Occurs mainly with very large leaks (e.g., tracheobronchial injury) or with underlying restrictive lung disease
Loss of effective tidal volume	Incomplete expansion of other lung areas Worsened ventilation–perfusion mismatching	Attempts to compensate with increased delivered tidal volume may increase leak and exacerbate problem
Inability to remove CO ₂	Acute respiratory acidosis	Unusual; life-threatening acidemia rare
Loss of positive end-expiratory pressure	Incomplete lung expansion Hypoxemia	May not be correctable in presence of large or multiple leaks
Pleural space infection	Infected secretions from airways pass through pleural space	Increases morbidity and ?mortality rates; ?additional source for sepsis
Factitious ventilator triggering	Ineffective ventilation	Caused by transmission of negative pressure from chest tube to central airways

Bronchopleural fistula is a relatively uncommon complication, even in institutions managing large numbers of patients with ARDS and other forms of acute respiratory failure. Of 1700 patients ventilated during a 4-year period at a major trauma center in the early 1980s, thirty-nine (2%) developed a bronchopleural air leak that persisted at least 24 hours after chest tube insertion.¹¹⁶ Postsurgical leaks, such as bronchial stump breakdown, were not included in this series. The leak varied in size in these patients from only intermittent bubbling into the chest drainage device to 900 mL/breath. Despite this, refractory hypercapnia causing arterial pH <7.30 occurred in only two patients, suggesting that inability to excrete CO₂ is very uncommon when bronchopleural fistula occurs as a complication during ventilator management.

Bronchopleural fistula occurred in two distinct settings in this study.¹¹⁶ It was either the result of direct lung injury (e.g., blunt chest trauma or instrumentation) occurring within 24 hours of the onset of mechanical ventilation, or a manifestation of spontaneous alveolar rupture considerably later in the clinical course. In the latter circumstance, most often seen in patients with ARDS, bronchopleural fistula first developed a mean of 13 days after the onset of positive-pressure ventilation. This case series, reported in the context of ventilator management three decades ago, remains the only published study of the incidence of persistent bronchopleural air leak as a feature of barotrauma.

As discussed earlier for barotrauma in general, it is not known whether bronchopleural fistula as a complication of ventilator support has become less common with the use of lung-protective ventilation. Likewise, the effect of this

complication on patient outcomes remains unknown. Development of a bronchopleural fistula appears to identify patients as having a poor prognosis, but whether the fistula plays an independent role remains to be established.^{116,117}

Management

General Principles

Of the clinical forms of barotrauma (see [Table 44-6](#)), pneumothorax and systemic air embolism present a threat to the life of the patient and thus require immediate treatment. All of the others should be considered primarily as signs, either of the severity of the pulmonary disease or of the possibility of inappropriate ventilator management. Extraalveolar air is not easily recognized at the bedside except in the case of subcutaneous emphysema and tension pneumothorax.

Nevertheless, there is often a temptation to initiate therapy directed at the barotrauma itself. More important, however, is therapy of the underlying primary disease. Certain ventilator adjustments, though, are appropriate in order to reduce the likelihood of further damage. These should include reducing PEEP, tidal volume, and minute ventilation to the lowest values compatible with acceptable patient support, and consideration of lung-protective ventilation and permissive hypercapnia as discussed elsewhere in this volume for reducing the likelihood of lung damage.

Pulmonary Interstitial Emphysema and Cystic Dilation

As mentioned earlier (see the section [Clinical Manifestations](#)), although pulmonary interstitial emphysema is an important intermediate step between alveolar disruption and clinical barotrauma, it is difficult to detect reliably on either physical exam or portable radiography obtained in the ICU. With the increasing use of CT in patients with ARDS and other settings of acute respiratory failure, however, these forms of barotrauma are increasingly recognized. Along with intraparenchymal air cyst formation, pulmonary interstitial emphysema is an important sign that alveolar overdistension is occurring, and that tension pneumothorax and other forms of barotrauma are more likely to occur. Their recognition is an indication to monitor the patient closely, and to modify the ventilator settings to reduce alveolar overdistension if this is clinically feasible. Aside from these measures, however, there is no specific treatment for these phenomena. Although CT-guided percutaneous catheter drainage of intraparenchymal air cysts has been reported,¹¹⁸ documentation of the clinical necessity of this is scant.

Subcutaneous Emphysema

In the great majority of cases, subcutaneous emphysema requires no treatment other than to treat the underlying lung pathology so that the leak will cease. Rarely, accumulation of massive amounts of air in the chest and abdominal walls can be physiologically important and require surgical decompression.^{119,120} Most accounts of such intervention provide little to convince the reader that the air was compromising vital organ function. If the air leak is massive and continuous, however, as can be seen in tracheal perforation as a complication of attempted intubation, urgent intervention may be warranted. In such instances, circulatory compromise, oliguria, and other manifestations of the abdominal compartment syndrome (such as increased bladder pressure) should be evident, and should improve dramatically upon release of the subcutaneous air.¹⁷

Pneumoperitoneum

Continued air leak resulting in massive spread of extraalveolar air during positive-pressure ventilation can lead to accumulation of air in many soft-tissue spaces. This air poses no threat to the tissues themselves, although in the case of pneumoperitoneum it can cause diagnostic confusion. Critically ill patients with barotrauma are also susceptible to

perforating stress ulcers, mesenteric ischemia, and other potential causes of free air in the peritoneal cavity, particularly if they have experienced a period of hypotension. Even in the presence of widespread subcutaneous emphysema and other forms of extraalveolar air, it may not be possible to exclude a surgical abdomen without laparotomy. One report documented the use of measured [oxygen](#) tension in the gas aspirated from the peritoneal cavity,¹⁰⁴ but this has yet to be confirmed by other investigators. Aside from the need to exclude an abdominal emergency unrelated to barotrauma, the presence of free air in the retroperitoneal tissue planes or peritoneal cavity generally poses no danger to the patient, and will be resorbed once the need for positive-pressure ventilation diminishes.

Systemic Air Embolism

The diagnosis of systemic air embolism is based on the clinical setting rather than on any specific test. Demonstration of air bubbles in the left atrium or left ventricle by echocardiography is highly suggestive of ongoing extraalveolar air entry, but CT and magnetic resonance imaging have not proven to be helpful in this condition.

No therapy is convincingly effective. Although hyperbaric [oxygen](#) therapy is often recommended, its efficacy has not been shown in controlled trials. Traditionally, it has been recommended that patients with air embolism be placed in the left lateral decubitus position to move the air collection out of the right ventricular outflow tract; evidence supporting this recommendation, however, is lacking and data from several animal studies suggest it may not be effective at improving patient hemodynamics.^{121–123} Administration of high concentrations of [oxygen](#) in order to raise partial pressure of arterial [oxygen](#) (PaO₂) and hasten bubble absorption has also been recommended,^{122,124} while reductions in PEEP, tidal volume, and minute ventilation should be carried out, if feasible, so as to reduce the likelihood of further air entry into the vasculature.

Pneumothorax

Because of the threat of tension pneumothorax, chest tube drainage is advisable anytime free pleural air is detected during positive-pressure ventilation. There is little reported experience in this setting with the small catheter drainage sets often used for treating uncomplicated primary spontaneous pneumothorax, and the use of a larger, standard chest tube and drainage system is advisable. The routine application of suction (e.g., 20 cm H₂O) is customary in the United States, although this is not the case in all areas of the world; chest tube suction should be applied or increased if the lung does not fully reinflate, although, as noted earlier, application of suction may perpetuate a bronchopleural fistula by maintaining a large gradient between the airways and pleural space. Once the lung is fully inflated and there is no residual air leak, the tube is placed to water seal and removed, usually after an additional day. No evidence supports the use of “prophylactic” thoracostomy tubes in patients at high risk for pneumothorax.

Bronchopleural Fistula

Diagnosis and Quantitation of Air Leak

In pneumothorax, once a chest tube is inserted and external suction applied, air is evacuated from the pleural space, as shown by bubbling through the water seal of the chest drainage device. If there is no persistent communication between the airways and the pleural space, this bubbling will cease once the lung is fully reinflated. Air may continue to leak into the pleural space for minutes to a few hours, but in most instances this soon stops. When it does not, and the bubbling continues for 24 hours or more, a bronchopleural air leak (bronchopleural fistula) is present. Most such leaks are small—a few bubbles through the water seal in synchrony with the inspiratory phase of the ventilator—although they may reach several hundred milliliters per breath.

Although precise quantitation of the air leak is rarely needed for clinical management, several methods for doing this have been described. The leaked volume can be estimated by subtracting the expired from the inspired tidal volume as long as the leak exceeds 100 to 200 mL/breath and the measurements are made at the same point in the ventilator circuit to avoid error secondary to compression in the inspiratory limb. Other techniques have been described in the literature but they are more difficult to set up, have lower overall clinical utility, and are, as a result, better suited for research purposes.^{125,126}

Although it is seldom necessary to quantitate the leak precisely for patient management, it is sometimes useful to determine whether the gas leaking into the pleural drainage unit originates in the patient's airways or from the room, as may occur around the chest tube insertion site or with breaks in the circuit between the patient and the drainage unit. Removal of the chest tube is unwise in the presence of a bronchopleural air leak, but may be indicated if there is an external leak, because the same circuit break that introduces air may also permit ingress of contaminating organisms. Most leaks from the patient's airways vary with the respiratory cycle, and stop or diminish during exhalation when intrathoracic pressure is lower. A more definitive test is to withdraw a sample of leaked gas from the connecting tubing under sterile conditions and to pass it through a capnometer; except in cases of proximal bronchial disruption, gas from the patient's airways will contain CO₂, whereas air from the room will not.

Bishop et al¹²⁷ collected leaked gas in nine patients with bronchopleural fistulas complicating ARDS. Total delivered minute ventilation in these patients ranged from 16.6 to 42.6 L/min (mean: 23.9 L/min), with 4% to 53% of each tidal volume lost through the fistula (mean: 25%). As the proportion of total minute ventilation lost through the leak increased, so did the fraction of the patients' total CO₂ production that exited through the chest tube.¹²⁷ In several patients, more CO₂ was excreted through the chest tube than through the endotracheal tube. Thus, at least in ARDS, gas leaked through a bronchopleural fistula has participated in gas exchange and is similar to gas exiting via the trachea.¹²⁷ This observation is consistent with the finding that unmanageable CO₂ retention is unusual in bronchopleural fistula,¹¹⁶ and suggests that respiratory acidosis in most patients with such a fistula is a manifestation of the severity of their underlying lung disease rather than of the air leak per se.

Measurements of oxygen consumption, CO₂ production, respiratory quotient, and resting energy expenditure can be erroneous in patients with significant bronchopleural air leaks because of the CO₂ lost through the chest tube.¹²⁸ Accurate measurement of CO₂ production in patients with ARDS requires collection of the leaked gas, although dead-space-to-tidal-volume ratio can be determined using the usual technique of endotracheally expired gas collection, because the proportion of CO₂ excreted via trachea and fistula are approximately equal.¹²⁸

General Measures

In general, management of the patient with a bronchopleural fistula is the same as if the fistula were not there, provided that a functioning chest tube and a pleural drainage system are in place. Table 44-10 lists a number of commonsense measures that should be taken to diminish the magnitude of the leak and the probability of further pulmonary parenchymal damage.

Table 44-10: Principles of Ventilator Management in the Patient with a Bronchopleural Fistula

Use the lowest number of mechanical breaths that permits acceptable alveolar ventilation (reduce both mean airway pressure and number of high-pressure breaths)

Wean the patient completely if possible

Partial ventilatory support may be preferable to total ventilatory support (e.g., pressure-support ventilation)

Avoid or correct respiratory alkalosis (to minimize minute ventilation)

Limit effective (returned) tidal volume to 6 to 8 mL/kg or less

Minimize inspiratory time

Keep inspiration-to-expiration ratio low (e.g., 1:2)

Use high inspiratory flow (e.g., 70 to 100 L/min)

Avoid end-inspiratory pause and inverse-ratio ventilation

Use low-compressible-volume ventilator circuit to minimize delivered tidal volume

Minimize both dialed-in and endogenous positive end-expiratory pressure

Use the least amount of chest tube suction that maintains lung inflation

Explore positional differences; avoid placing the patient in positions that exacerbate leak

Treat bronchospasm and other causes of expiratory airflow obstruction

Consider specific or unconventional measures (e.g., independent lung ventilation, endobronchial measures) if the patient remains unstable or develops clinically harmful, uncorrectable respiratory acidosis despite the above measures

Treat the underlying cause of respiratory failure, maintaining nutritional and other support, with the goal of discontinuing mechanical ventilation as soon as possible

Weaning the patient from positive-pressure ventilation altogether would be optimal from the standpoint of decreasing the leak. When this is not feasible, a ventilator strategy should be selected that minimizes both minute ventilation and mean intrathoracic pressure. Respiratory alkalosis should be avoided, and consideration should be given to permissive hypercapnia in patients with very low lung–chest wall compliance and high minute ventilation requirements. Because the “lost” volume participates in gas exchange, delivered tidal volume should not be increased beyond a certain point to “chase” the air leak.

Positive end-expiratory pressure (both dialed-in and endogenous) and pulmonary hyperinflation should be minimized. In a study using older-generation ICU ventilators and circuits, the use of high inspiratory flows and low-compressible-volume, low-compliance ventilator tubing independently and additively decreased auto-PEEP,¹²⁹ although whether these measures are necessary with current-generation ventilators and circuits is uncertain. High inspiration-to-expiration ratios and end-inspiratory hold should be avoided. Vigorous therapy for bronchospasm and obstructing airway secretions should be pursued.

Specific Measures

As stated previously, the role of a bronchopleural fistula in the overall prognosis of acute respiratory failure has not been established, and available data suggest that the great majority of such fistulae have little impact on gas exchange. Of all the clinical forms of barotrauma, however, this is the one that has been the subject of the most published reports of novel approaches to therapy. Most of the literature consists of descriptions of techniques for decreasing the size of the leak, and available clinical reports consist almost exclusively of anecdotal descriptions of immediate and short-term physiologic changes that lack both longer-term outcome data and convincing evidence that “conventional” ventilator management

had been inadequate. The subjects of most of these reports have been critically ill patients, many of whom have died in spite of the short-term physiologic improvements documented under the reported intervention.

Several reports have described the use of independent lung ventilation via double-lumen endotracheal tube and two ventilators in patients with bronchopleural fistula.^{130–136} The reported experience with this strategy is limited to about ten patients, although the technique has no doubt been used in other unpublished cases. As with the use of this ventilator technique in other settings, improved gas exchange has generally been achieved, but in nearly all the reported cases, the necessity of switching to this complicated and expensive technique is unconvincingly described, a problem shared with many of the other strategies described further below. Placement and maintenance of a double-lumen endotracheal tube in a critically ill patient require considerable expertise, and the small lumens of these tubes make bronchial hygiene difficult in patients with significant secretions.

Many older reports documented the use of high-frequency jet ventilation (HFJV) in bronchopleural fistula,^{137–143} but enthusiasm has waned and there are few new reports of the use of this technique in the last two decades or more.^{144,145} Clinical experience with HFJV in patients without underlying lung disease, as in traumatic bronchial disruption or during tracheobronchial surgery, has generally been positive.^{143,146} When bronchopleural fistula, however, occurs as a manifestation of barotrauma complicating severe diffuse pulmonary disease, both short-term and outcome results have been discouraging. In one study of seven patients with severe ARDS and large air leaks, when HFJV was adjusted to provide the same mean airway pressures as resulted from conventional ventilation, oxygenation and/or effective alveolar ventilation deteriorated in every patient, and the size of the leak increased in five of seven patients.¹⁴⁷ Because of this experience, the use of HFJV as treatment for bronchopleural fistula complicating severe acute respiratory failure has been abandoned in the authors' institution. High-frequency oscillatory ventilation has been reported in two cases of bronchopleural fistula,¹⁴⁸ but whether this technique offers any advantage over adjustments in conventional ventilation remains to be seen.

Several approaches have been reported for decreasing the size of the air leak through manipulation of the pleural drainage system.^{149–152} The total patient population in which these approaches have been reported is very small, and no new reports of their successful use have appeared in more than 20 years. Their application becomes progressively more difficult as the number of chest tubes increases in a given patient, and incomplete lung expansion on the side of the leak is a common problem.

Commercial chest drainage units vary considerably in their ability to handle air leaks. In one bench study, the maximum leak flow that four widely used commercial chest drainage units could accommodate varied from 5.8 to 35.5 L/min.¹⁵³ In a subsequent study using an animal model of bronchopleural fistula, two older but commonly used devices became ineffective at leak flows as low as 4 to 5 L/min because of increased resistance.¹⁵⁴ A freestanding suction pump independent of wall suction (Emerson suction pump) consistently demonstrated the highest capacity to evacuate leaked air in these studies, and when flow through a bronchopleural fistula exceeds 4 to 5 L/min the use of such a device is recommended. We are unaware of studies evaluating the performance of the current generation of pleural drainage systems with respect to evacuating large air leaks.

Several approaches have been reported for plugging or otherwise sealing bronchopleural air leaks using the flexible or rigid bronchoscope,^{144,155–165} including the use of endobronchial one-way valves^{166,167} and expandable stents.¹⁶⁸ Very few of the patients comprising these case reports and small series, however, have had bronchopleural fistulas in the

setting of ventilator-associated barotrauma.^{156,162,163} Most of the reported experience is with bronchopleural fistulas following lung resection.¹⁶⁹

Although it is tempting to think that the ultimate therapy for bronchopleural fistula would be direct surgical closure, this is seldom possible for technical reasons in settings other than acute traumatic tracheobronchial disruption. If the fistula is localized and associated with a necrotizing pneumonia, resection of the affected lobe may be feasible. A bronchopleural air leak following lung biopsy or other procedure, in a patient without pneumonia or acute respiratory failure, may be amenable to surgical repair by thoracoscopy or thoracotomy. In most instances of barotrauma, however, the air leak is a manifestation of the severity of the underlying pulmonary disease, and direct suture or cautery of the leak or leaks is simply not technically possible.

The clinician seeking the best way to manage a patient with bronchopleural fistula complicating positive-pressure ventilation is poorly served by the existing experimental literature, which consists largely of technical descriptions and anecdotal reports of immediate changes in leak size and arterial blood-gas values, and is virtually devoid of evidence that the therapies described affect patient outcome. There is an understandable desire to intervene when something so obvious as a bronchopleural fistula develops during the course of ARDS or other form of severe acute respiratory failure. Yet, except for the unusual circumstance of inability to reinflate the affected lung, evidence for harm by the air leak per se is strikingly lacking, and all the reported techniques for diminishing it have the potential for serious harm to the patient.

The clinician should approach any unconventional therapeutic measure with caution, realizing that it constitutes experimental, unproven, and potentially harmful therapy for a condition that may be more a sign of an adverse prognosis than its mechanism. [Table 44-11](#) summarizes a number of principles for the approach to managing patients with bronchopleural fistula complicating severe acute respiratory failure, taken mainly from clinical experience in an area whose scientific basis remains woefully inadequate.

Table 44-11: Eight General Observations About Managing Bronchopleural Fistula in the Patients with Acute Respiratory Distress Syndrome

1. Bronchopleural fistula in the patient with ARDS is a manifestation of the severity of the underlying disease, perhaps initiated or exacerbated by its management, that will not resolve until the ARDS resolves or improves.
2. Bronchopleural fistula in ARDS is physiologically significant in only a minority of cases (perhaps 10%), even in the presence of hypercapnia.
3. Reducing the leak size per se typically has little effect on gas exchange as measured by arterial blood gases.
4. Measures directed at decreasing the size of the leak will generally prove unsuccessful until the underlying lung injury improves.
5. When the ARDS improves, the bronchopleural fistula will nearly always improve without specific therapy.
6. None of the specific measures (e.g., high-frequency ventilation, bronchoscopic interventions, etc.) discussed in the text has been shown to affect patient outcome.
7. Patients almost never die *of* bronchopleural fistula. They die *with* bronchopleural fistula, usually *of* multiple organ failure and occasionally of gas exchange failure because of ARDS.
8. Careful attention to general management principles ([Table 44-10](#)) is more important than any specific measure to decrease the leak.

Abbreviation: ARDS, acute respiratory distress syndrome.

Prevention

The degree to which all forms of clinical barotrauma are iatrogenic, as a result of how ventilator support is used in severe respiratory failure rather than primarily of the underlying pathology, remains unclear. Nonetheless, from the material discussed in this chapter several general principles emerge that are reasonable if not scientifically proven to reduce the likelihood of barotrauma during mechanical ventilation. [Table 44-12](#) summarizes these principles.

Table 44-12: Practical Measures to Reduce the Likelihood of Barotrauma during Mechanical Ventilation

- Use small tidal volumes (6 mL or less per kilogram predicted body weight) and other aspects of lung-protective ventilation in patients with ARDS or obstructive lung disease, particularly in the presence of PEEP (dialed-in or endogenous)
- Avoid hyperventilation except where deliberately employed for specific indications
- Use PEEP cautiously in patients at increased risk for alveolar rupture, including those with:
 - ARDS
 - Any unilateral, patchy, or cavitary lung disease
 - Nosocomial pneumonia or sepsis
 - Obstructive lung disease (e.g., COPD or asthma)
 - Cystic lung disease (cystic fibrosis, *Pneumocystis jiroveci* pneumonia, lymphangioleiomyomatosis)
- Monitor respiratory system compliance as PEEP is applied or increased; decrease PEEP if compliance falls with increasing levels
- Monitor all ventilated patients for auto-PEEP; take specific measures to reduce auto-PEEP if its presence could be harmful to the patient:
 - Reduced minute ventilation, with normocapnia or hypercapnia
 - High inspiratory flow (e.g., 70 to 100 L/min)
 - Low-compressible-volume, low-compliance ventilator circuit
- Use extreme care in high-risk patients when placing subclavian or internal jugular lines or performing thoracentesis. Where available, use ultrasound guidance
- Take special care to avoid nosocomial pneumonia:
 - Practice good hand cleansing and other infection control techniques
 - Avoid circuit interruption

Abbreviations: ARDS, acute respiratory distress syndrome; COPD, chronic obstructive pulmonary disease; PEEP, positive end-expiratory pressure.

The large (e.g., 12 mL/kg) tidal volumes traditionally used in ventilating adult patients originated many years ago in studies of individuals with normal lungs who were undergoing anesthesia. These tidal volumes are inappropriate for patients with diffuse lung disease, either restrictive (e.g., ARDS) or obstructive (e.g., chronic obstructive pulmonary disease or severe asthma). For such patients considerably smaller tidal volumes (e.g., 6 mL/kg, proportioned to predicted body weight) should be used, especially when PEEP is present with its attendant risk for local or overall pulmonary hyperinflation. Other patients should be ventilated with tidal volumes not exceeding 8 to 10 mL/kg predicted body weight.

PEEP, whether externally applied or endogenous, increases the likelihood of pulmonary hyperinflation, particularly when the lungs are heterogeneously involved, as is now recognized to be the case in ARDS. Insofar as hyperinflation predisposes to alveolar disruption, PEEP should be used cautiously in ARDS, and auto-PEEP should be sought in all ventilated patients and minimized when present. Because of the increased propensity of patients with ARDS to develop barotrauma late in the course of this disorder, and the apparent association of alveolar rupture with ongoing pulmonary infection and inflammation, scrupulous attention to infection control measures in all aspects of the care of such patients seems especially important.

Important Unknowns

Surprisingly little is known about barotrauma complicating mechanical ventilation in terms of pathogenesis, clinical implications, optimum management, and prevention. The literature on this topic consists mainly of anecdotal observations and reports of interventions, the latter generally incompletely documented and at best with short-term physiologic endpoints. Although tension pneumothorax, air embolism, and, rarely, pneumomediastinum and subcutaneous emphysema, produce potential threats to life, whether the other forms of barotrauma discussed in this chapter exert independent effects on morbidity, mortality, or the natural history of acute respiratory failure is not known. Even bronchopleural fistula, one of the more dramatic forms of clinical barotrauma, has not been shown to affect any patient-relevant outcome. The trend toward use of smaller tidal volumes, particularly in patients with acute lung injury or obstructive lung disease, would be expected to reduce the incidence of barotrauma, although to date the available data has not shown this to be the case.

The Future

If the use of lower inflating pressures and distending volumes during mechanical ventilation, which is becoming widespread largely as a result of the demonstration that such measures reduce mortality in patients with acute lung injury, also reduces the risk of alveolar disruption, then clinical barotrauma should become less frequent. The lack of evidence that special devices or unconventional approaches to ventilator support reduce the incidence of barotrauma or hasten its resolution once present may lead to their being used less often in patient management, which may lessen costs and the risk for attendant complications. In view of the lack of evidence that specific measures impact the severity or natural course of barotrauma, focusing on prevention through less injurious ventilatory support and better general care during critical illness is likely to be more beneficial than the continued search for specific interventions for its management.

Summary and Conclusion

Part of a spectrum of complications occurring during mechanical ventilation that includes ventilator-induced lung injury, extraalveolar air can be the result of alveolar disruption from overdistension or be caused by several other mechanisms. As a possible result of the ventilator settings used, barotrauma is always of clinical concern. Among its several clinical forms, however, those of greatest importance are systemic gas embolism and pneumothorax, as these can be fatal. Rarely, pneumopericardium and subcutaneous emphysema can also create a threat to life, and pneumoperitoneum poses a dilemma with respect to the possibility of a ruptured abdominal viscus. Bronchopleural fistula is an indicator of pathology that needs attention, but is usually not physiologically important and seldom requires specific measures targeted at the leak site. Barotrauma from alveolar rupture tends to occur in severely ill patients, and whether this complication independently contributes to mortality or morbidity remains unknown. Prevention of barotrauma, and management once it occurs, consist primarily of practicing sound general ventilator management, including limiting distending volumes and static airway pressures, to stop or reduce the passage of air into the bronchovascular sheaths and elsewhere.

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